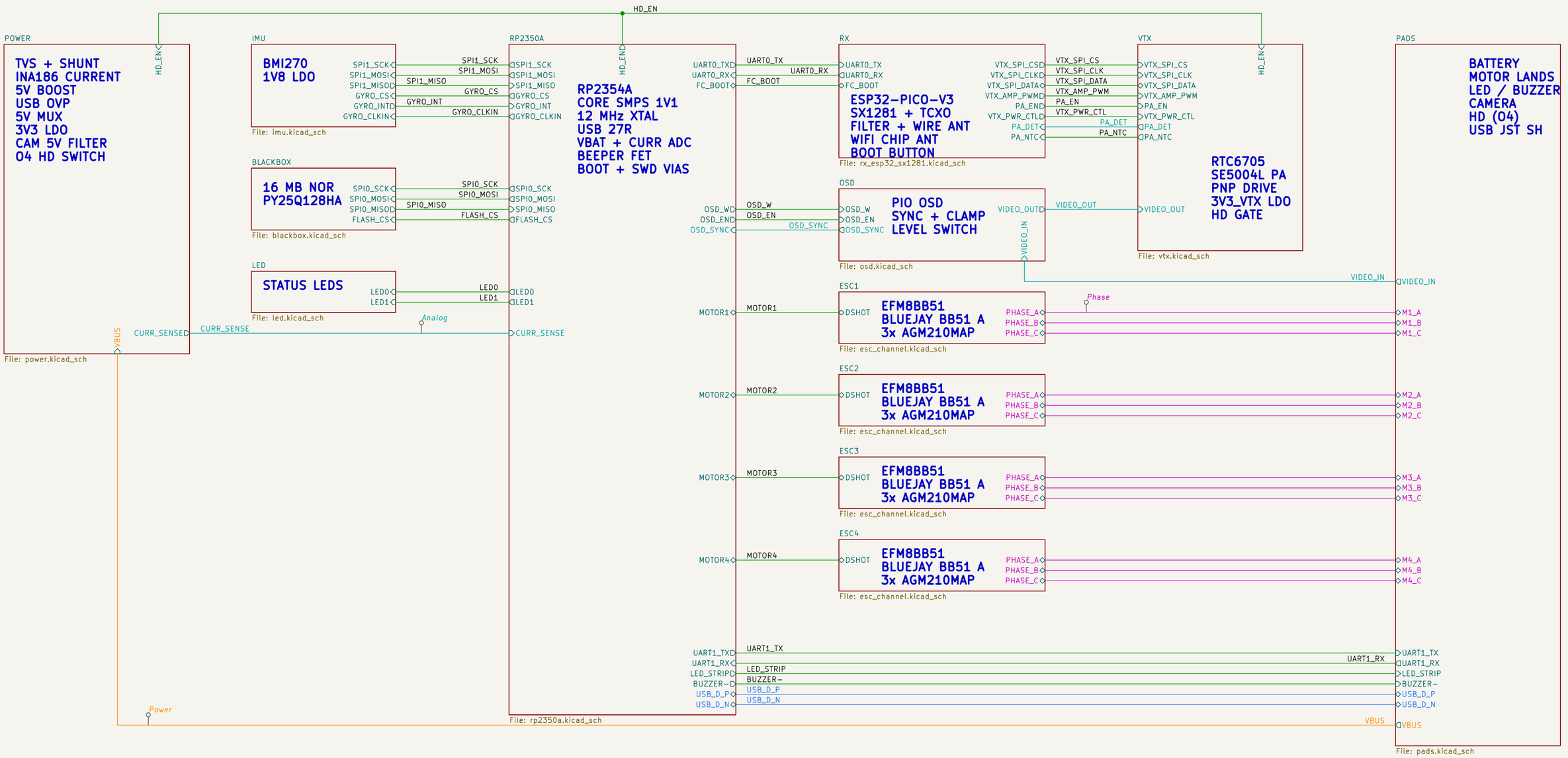


POWER TABLE

Rail	Source	Nominal	Budget (spec 5)	Feeds
+BATT_IN	battery pads J2 / J3 (1S)	3.0–4.35 V	4x ESC + 2.0 A board	TVS, pad bulk, shunt R1
+BATT	shunt R1 0.5 mOhm	cell	ESC + boost in 1.4–3.0 A	ESC1–4, boost, O4 EN, VBAT ADC, PA_VCC
/VBUS	USB J31 pin 4 (hier. net)	5 V	0.42 A, USB only	U24 OVP
+5V_USB	U24 SGM40661 OVP	5 V	as VBUS	U3 VIN1, PR1 divider
+5V_BST	U2 TPS61022 boost	5.15 V	0.70 A analog / 1.48 A HD (BEC 1.5 A cont.)	U3 VIN2, U5 O4 switch
+5V	U3 TPS2116 mux	5 V	0.70 A analog, about 0.25 A HD	3V3 LDOs, CAM filter, 5V pads
+5V_CAM	FB1 ferrite from +5V	5 V	0.12 A	camera pad + plug
+5V_HD	U5 TPS22810 (HD_EN, cell)	5 V	1.20 A O4 Lite	VHD pad
+3.3V	U4 LP5912–3.3	3.3 V	0.20 A (0.48 A Wi–Fi bench)	FC, IMU, NOR, OSD, RX, LEDs, INA186
+3.3V_VTX	U21 LP5912–3.3 (off in HD)	3.3 V	0.10 A typ, about 0.17 A at max PA	RTC6705, PA drive, PA ref
+1.8V_GYRO	U12 TPS7A2018	1.8 V	< 1 mA	BMI270
+1.1V	RP2354A core SMPS	1.1 V	internal	RP2354A DVDD
PA_VCC	R67 selector, +BATT default	cell	0.65 A at 27 dBm	SE5004L PA (local, VTX)
PA_VREF	U22 LP5907–2.85 (EN = PA_EN, off in HD)	2.85 V	about 10 mA	SE5004L VREF/EN (local, VTX)
GND	battery –	0 V	–	all sheets

Generated from hardware/sch_contract.json by hardware/tools/sch_root.py: do not edit by hand. Root wires carry the project-wide net names (root label = net /NAME). Motor phases are named by the ESC sheets (/ESCn/PHASE_x). Rails are power symbols (table); VBUS is a hierarchical net. ESC1–ESC4 drive M1 rear right, M2 front right, M3 rear left, M4 front left (Betaflight quad–X); FC pin map: research/PINMAP.md.

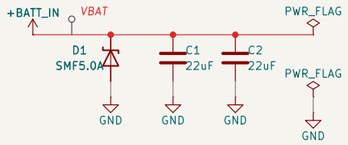
OpenDrone

Sheet: /
File: OpenAIO–Whoop.kicad_sch

Title: OpenAIO–Whoop overview (1S whoop AIO)

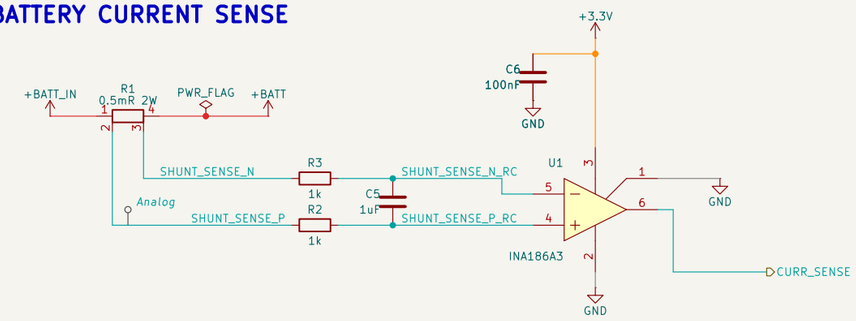
Size: A3	Date: 2026–10–09	**Rev: rev1**
KiCad E.D.A. 10.0.6		Id: 1/14

BATTERY INPUT AND TVS



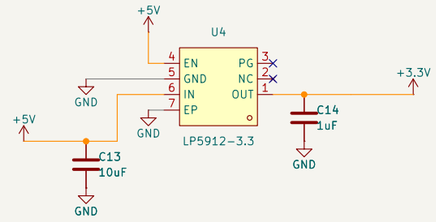
Source: OpenAIO/hardware/OpenAIO.kicad_sch (battery entry), adapted.
+BATT_IN = battery pads J2 (+) / J3 (−) on the PADS sheet, 1S LiPo/LiHV 3.0–4.35 V; PWR_FLAG here (the pads are passive). No reverse-polarity FET (spec 4.1: user-fitted pigtail, risk in README).
D1 SMF5.0A: VBR 6.40–7.00 V (10 mA), VC 9.2 V at 21.7 A, 200 W 10/1000 us: takes one motor's regen energy (~0.29 J per rated pulse) on a battery disconnect; fallback SMAJ5.0A (SMA, 400 W) if V3b exceeds it. It does not hold the 5.5–7 V parts in rating (EFM8 VDD on the cell; TPS61022 VIN 7 V abs max, and its pass-through carries a surge to +5V_BST = TPS2116 VIN2, 6 V abs max): Matrix-parity risk, spec 15.4.
C1/C2 22 uF 16 V X5R 0603 at the pads: 9–13 uF effective each at 4.35 V. Hot plug of a 4.35 V pack can overshoot to 5.0–5.4 V (V3c); above the TPS61022 4.8 V VIN limit for a start-up with VOUT pre-biased < 0.7 V (DS 6.3), which is exactly hot plug (EN = VIN, +5V_BST at 0 V); V3c scopes VIN with the +5V_BST start. Damping fallback ~100 uF polymer at the pads; PE-8 waits for V3c.

BATTERY CURRENT SENSE



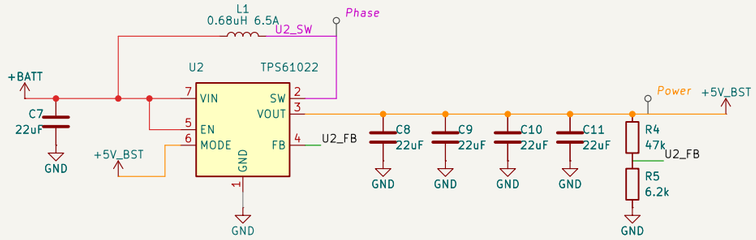
Source: OpenAIO/hardware/OpenAIO.kicad_sch (shunt + INA186A3, root sheet); input RC moved to 0201 and its two common-mode caps deleted (D52 BR-07); the output RC (1k/100n) sits at the ADC pin (RP2350A sheet, R47/C60).
P4-9: R1 HCS1206 0.5 mOhm on the four-pad Kelvin net-tie land (pad groups 1–2 and 3–4), SHUNT_SENSE_P = pad 2 (battery side). SHUNT_SENSE_N = pad 3: each holds only its inner sense pad and R2/R3; SHUNT_SENSE_P_RC/N_RC hold R2/R3, C5 and U1 (netlist checked); all four match the Analog and Kelvin DRU patterns. Route the pair on L1 or L6, never In2/In3 (DRU), class Analog. Everything that draws current taps +BATT after R1.
R2/R3 1k + C5 1 uF differential: fc = 1/(2 pi x 2 kOhm x 1 uF) = 80 Hz. IIB <= 3 nA x 1 kOhm = 3 uV (6 mA), VOS <= 50 uV (0.1 A).
U1 INA186A3, gain 100: 0.5 mOhm x 100 = 50 mV/A. Betaflight ibata_scale 500: output to VS − 40 mV gives about 65 A full scale on +3.3V; VCM −0.2..40 V covers the cell; REF to GND (unidirectional); C6 100 nF at VS. Shunt loss 0.2 W at 20 A. Scale checked on every prototype (V-plan).

3.3 V LDO FOR FC + RX



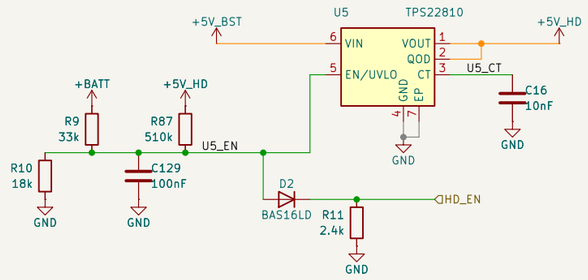
Source: OpenFC-Lite-Mini/hardware/power.kicad_sch ('3.3V 0.5A MAIN LDO', LP5912-3.3), with the OpenDrone LP5912-3.3DRVR symbol (pins in package order: IN/EN on the left here, EN tied to +5V, no EN shared with another rail; U21 on VTX has its own EN). PG and NC open; EP to GND for heat (TI: GND or open).
P4-17 / BR-14: C13 10 uF X6S 0402 is the one CIN of U4 and U21 (VTX sheet): place it between the two IN pins (~2.4 mm apart). 2.0 uF effective at 5 V (Murata curve) > the 0.7 uF LP5912 minimum.
P4-12, +3.3V COUT at 3.3 V: C82 10 uF 3.5 uF (Murata GRM155C80J106ME11 DC-bias curve). C58 4.7 uF 0402 X5R -2.6 uF, 3x 1 uF 0201 X5R (C14, C61, C92) -0.45 uF each, 15x 100 nF X6S 0201 -0.085 uF each (last three typical, I): 8.7 uF typical, 7.1 uF at −20 %, 10.3 uF only with every part at +20 % -> inside TI's 0.7–10 uF (SNVSA77D 7.6) but for that +3 % corner, and most of it is remote (5–30 mm of trace), which the LP5912 allows. Lever if a V-bench load step rings: PS-O6 (delete C92). +3.3V_VTX is summed on the VTX sheet.
Load 0.20 A: (5.15 − 3.3) V x 0.2 A = 0.37 W, +26 K (RthJA 71.2 K/W). U4 OUT (project-lib LP5912 copy, power_out pin) drives +3.3V: no PWR_FLAG.

5 V BOOST TPS61022



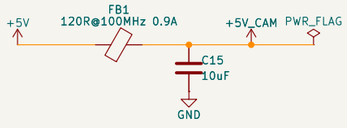
New block from TI SLVSDX7D (no sibling uses the TPS61022); look matched to the OpenFC-Lite-Mini buck blocks.
P4-7: VOUT = 0.600 V x (1 + 47k / 6.2k) = 5.148 V; with VFB 585–615 mV and 1 % resistors 4.93–5.37 V. R5 6.2k < 300 kOhm (TI). EN tied to VIN (+BATT) and nothing else (BR-11): same −0.3..7 V abs max, VEN_H <= 1.2 V, on whenever the cell is above UVLO. MODE = VOUT (+5V_BST): forced PWM, VMODE_H <= 1.2 V valid once VOUT > 2.2 V (start-up pass-through holds VOUT near VIN).
C7 22 uF CIN at VIN/GND. C8–C11 4x 22 uF X6S 0603: ~20–28 uF effective at 5.15 V, inside TI's 10–50 uF; TI 6.3 asks >= 10 uF effective up to 1.5 A and >= 20 uF from 1.5 to 3 A, met only typically, so the published BEC is 1.5 A continuous (D79; HD load 1.43–1.48 A, LED strip limited in HD mode; V1 load step at 1.7 A). Feed-forward cap (TI 8.2.2.4: above 40 uF effective, fFFZ 2 kHz = 1.7 nF across R4): C8–C11 give 20–28 uF, HD mode adds C12/C13 (~6 uF via the mux) and the O4 input (~47 uF, U5 note): none fitted (lean). V1b steps the load in HD with a real O4, 1.5–2.2 nF across R4 only if it rings. L1 0.68 uH, Isat 6.5 A, 17 mOhm; valley limit 6.5–10 A; converter capability 1.67/1.82/1.98 A at 2.8/3.0/3.2 V (spec 4.2). Passes VIN to VOUT when VIN > VOUT (surge path, spec 4.1).
Layout (TI 10.1, D92): C8–C11 VOUT/GND at pins 3/1 (hot loop SW–VOUT–Cout–GND smallest), C7 at pins 7/1, SW copper minimal, R4/R5 at FB pin 4 away from SW; ground return: SHEET NOTES (D92).

O4 LITE HD SUPPLY SWITCH TPS22810



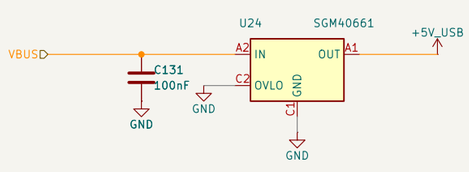
New block from TI SLVSDH0C (D15, D34; D53 BR-10/22/24). U5 = TPS22810DRVR WSON-6 2x2 (D83, P4 r4: smallest package, 3 A at TA 65 C vs 2 A DBV, RthJA 74.6 vs 182 C/W; same die and EN/UVLO figures; pins 1 VOUT 2 QOD 3 CT 4 GND 5 EN/UVLO 6 VIN, EP 7 to GND as on TI's EVM, SLVUAY7B Fig. 1).
P4-2 / BR-10: U5 VIN on +5V_BST, ahead of the mux: USB never reaches the O4. No separate CIN (C17 deleted, P4 critique): the boost outputs C8–C11 (20–28 uF effective) are U5's input capacitor (TI 10.3: CIN optional, 1 uF usually sufficient): P5 condition: U5 VIN within 3 mm of C8–C11 or U3 VIN2 on the +5V_BST copper. C16 CT 10 nF: ~4.7 V/ms, 0.2 A inrush per 47 uF. QOD tied to VOUT: the internal RPD empties +5V_HD when off.
U5_EN (bom_plan O4_EN) holds exactly R9 33k / R10 18k from +BATT, R87 510k from +5V_HD, C129 100 nF, D2 anode and U5 EN/UVLO pin 5 (netlist checked). D2 BAS16LD cathode + R11 2.4k pull-down on HD_EN (GPIO27, PINI01).
P4-1 (VENR 1.13–1.30 V, VENF 1.08–1.18 V, +5V_HD 4.93–5.37 V): EN = 0.345 x cell (+0.0223 x +5V_HD when on). Nominal R: on at 3.27–3.77 V, shed at 2.78–3.10 V; 1 % R: on 3.23–3.82 V, shed 2.74–3.14 V; resistor hysteresis 0.327–0.340 V >= the 0.31 V cell rebound.
HD_EN (GPIO19) low or undriven: EN <= VF 0.715 V + 66 uA x 2.4k = 0.87 V < VENF 1.08 V, also with +5V_HD still up. HD_EN high: D2 reverse >= 1.7 V; 1 uA hot leakage moves EN 11 mV (shed 0.03 V); R11 draws 1.4 mA.
Filter: 11.4 kOhm x 100 nF = 1.14 ms (140 Hz): the 96 kHz ESC ripple (0.21 V p-p at EN) falls to 0.3 mV. D2 + R11 discharge it below VENF in 0.25 ms; turn-on from the clamp ~1.6 ms at 4.2 V. USB only (back-fed 2.5–2.8 V + 0.3 V ripple): EN <= 1.07 V < VENR. EN <= 1.62 V << 18 V abs max.

CAMERA 5 V FILTER



Source: OpenFC-Lite-Mini/hardware/power.kicad_sch (camera 5 V feed), with the commons 120 Ohm 0201 bead (blocks/LDO_IMU_TPS7A2033).
FB1 BLM03PX121SN1D: 120 Ohm at 100 MHz, 0.9 A, 160 mOhm: the 0.12 A camera drops 19 mV; it keeps the boost's 1 MHz and the camera's own switching apart. C15 10 uF X6S after the bead. +5V_CAM feeds the CAM pad J24 and the camera plug J32 (PADS); PWR_FLAG because FB1 is passive. Fallback BLM03PX220SN1D (1.45 A).

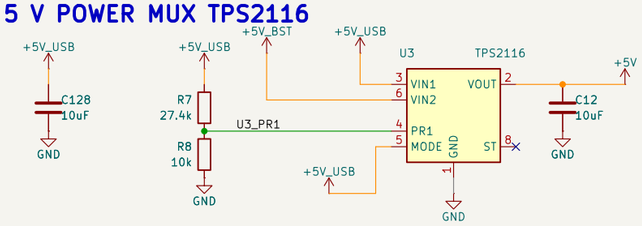
USB INLET OVP SGM40661



New block from the SGMICRO SGM40659/60/61 datasheet REV.A (D57).
IN = VBUS (J31 pin 4, PADS), OUT = +5V_USB. OVLO (C2) to GND selects the internal threshold: 5.75–6.12 V rising, 5.45–5.66 V falling. IN 28 V abs max, 2.7–20 V recommended (feature list: 40 V surge immunity); UVLO 2.4 V. 15 ms debounce, then a 2 ms soft turn-on: a cable LC ring (simulated <= 10.8 V at IN) never reaches TPS2116 VIN1 (6 V abs max), and the soft turn-on is the VBUS surge limit USB asks for above 10 uF downstream.
C131 100 nF X6S 25 V at the IN balls

(datasheet CIN 0.1 uF; also takes ESD charge). COUT = C128 10 uF at U3 VIN1 (<= 100 uF allowed). RON 27 / 45 mOhm: 11–19 mV at 0.42 A. TA −40..85 C.
Open: a non-compliant source at 5.5–6.12 V would pass before the OVLO trips (TPS2116 VIN1 5.5 V recommended, 6 V abs max): USB sources are <= 5.25 V (5.5 V with PD).

5 V POWER MUX TPS2116



New block from TI SLVSFG1A, priority mode (D24). C128 = the VIN1 cap: place it at U3 pin 3 (TI 9–10).
MODE tied to VIN1 (+5V_USB). PR1 = VIN1 x 10k / 37.4k: USB is selected above VREF x 3.74 = 3.44–4.04 V (VREF 0.92–1.08 V; 3.37–4.10 V with 1 % R), +5V_BST (VIN2) otherwise. ST (open drain) unused.
P4-2: +5V_BST holds only U2 VOUT/MODE, C8–C11, R4, U3 VIN2 and U5 VIN (netlist checked). USB selected: VIN2 stays off (truth table 'VIN2 = X') and both channels block reverse current, so USB never feeds +5V_BST or the O4.
P4-3: C128 10 uF X6S 0402 at VIN1 (TI CIN 1 uF): VIN1/MODE/PR1 peaks are bounded by U24. Unplug with a cell: +5V follows VBUS to the PR1 threshold, then the 8 us break-before-make gap: dV = 0.7 A x 8 us / (C12 + C13 + C15, 3.4 uF each at 3.4 V, Murata curve) = 0.55 V -> +5V >= 3.19 V typical: worst corner (3.44 V, −20 % caps) 2.75 V. +3.3V stays >= 3.07 V: the LP5912 blocks reverse current and its >= 7 uF output carries 0.2 A for 8 us. V1 case (b) scopes it.
C12 10 uF at VOUT: +5V hold-up. Mux 40/60 mOhm: <= 0.04 V at 0.70 A analog; HD about 0.25 A (the O4 is on +5V_BST ahead of the mux, D53 BR-10).
TI 7.3.4 (clamp VOUT or about 100 uF where RCB occurs): RCB happens by design at USB plug/unplug with a cell; considered deviation, no part (short +5V copper, near-zero current at the 42–70 mV trip). V1: VIN1/VOUT <= 6.0 V.

SHEET NOTES

POWER sheet, generated with hardware/tools/sch_build.py + sch_root.py (P4 power shard); do not edit by hand.
Rails are power symbols (net = Value): +BATT_IN, +BATT, +5V_BST, +5V_USB, +5V, +5V_CAM, +5V_HD, +3.3V, GND. Sheet pins: HD_EN (in), CURR_SENSE (out), VBUS (in). Local nets: SHUNT_SENSE_P/N(_RC), U2_SW, U2_FB, U3_PR1, U5_EN.
Closed here: P4-1, P4-2, P4-3, P4-7, P4-9, P4-12 (+3.3V), P4-17 (BR-14 C13), P4-19 (C30 deleted): R11 is the HD_EN (GPIO19 since the D111 map) pull-down P4-4 counts on; P4-13 is void (D69: C2, C5, C14 stay X5R, sized on DC bias). Values: research/DESIGN-SPEC.md 4.1, 4.2, 15.5; P4 hand calculations reproduced in the shard notes.
P4 critique round 1 (D77, D108): C30 (ESC3 EFM8 VDD bulk) deleted: each ESC has one 22 uF 6.3 V X6S 0603 hot-loop cap per AGM210MAP on its own sheet (D108, was 10 uF 0402), and ESC3's slot-M cap C35 sits within 4 mm of U8 VDD pin 4 (BR-05 as amended, 100 nF C31 at the pin). C17 (U5 CIN) deleted: C8–C11 serve, P5 placement condition above.
D92 5 V boost ground return (U2, L1 and C7–C11 on F in the winning placement V4_B4: component layer L1, adjacent GND plane L2). The pulsed current is on the output side (SW -> high-side FET -> VOUT pin 3 -> C8–C11 -> GND -> low-side FET at GND pin 1), so C8–C11 GND, C7 GND and U2 pin 1 form one component-layer copper island and the loop closes on it: no via inside the loop, no copper neck between the caps and pin 1. The island joins the GND planes (L2/L5) through its own GND vias at the C7 (input) end, outside the loop: they carry the battery-side return, 2.9–3.6 A DC at 1.5–2.0 A out from a 3.0–3.2 V cell (eff. 0.9), so >= 4 vias at the FLOORPLAN 1 A per via. The adjacent plane stays unbroken under the loop, SW node and L1 (L2 stays solid, D85). R5 returns to pin 1 on its own trace, not into the C8–C11 copper; R4 senses VOUT at C8–C11. Keep the island out of the J3 to ESC-cell return corridor and the camera/OSD/VTX video ground, and U2_SW / L1 away from the SHUNT_SENSE / CURR_SENSE traces, gyro, crystals and antennas.

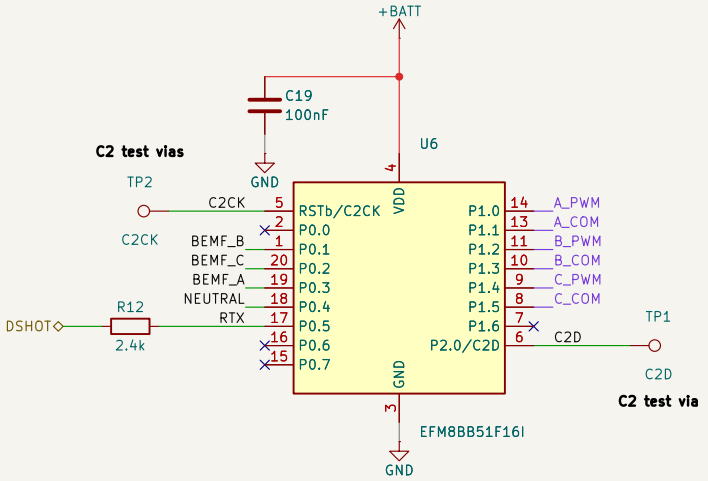
OpenDrone

Sheet: /POWER/
File: power.kicad_sch

Title: Power: input, current sense, 5 V, 3.3 V, O4

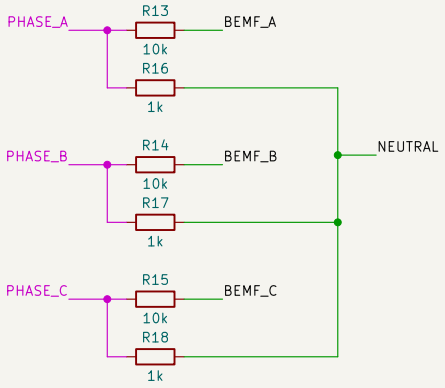
Size: A3	Date: 2026-10-09	**Rev: rev1**
KiCad E.D.A. 10.0.6		Id: 2/14

ESC MCU: EFM8BB51, Bluejay BB51 layout A



Pin use = Bluejay src/Layouts/BB51/A.inc (0368d11): P0.1 B_Mux, P0.2 C_Mux, P0.3 A_Mux, P0.4 V_Mux (neutral), P0.5 RTX; P1.0/P1.1 A_Pwm/A_Com, P1.2/P1.3 B, P1.4/P1.5 C (PWM active high = N gate, COM active low = P gate, all push-pull); P2.0 DebugPin = C2D. Pin numbers: EFM8BB51 data sheet Rev 1.0 Table 6.1. P0.0, P0.6, P0.7, P1.6 unused: no-connect (PINMAP 7.1).
VDD pin 4 straight on +BATT, no series R (it must track the P sources); C19 100 nF X6S at pin 4 / GND pin 3. The data sheet's 1 uF (Fig. 5.1) is the phase B hot-loop cap C23 22 uF X6S 0603 (D108; about 7–10 uF effective at 4.2 V); BR-05 as amended (P4 critique): within 4 mm of pin 4 through the +BATT plane, 100 nF at the pin (v3 cell site 3.4–3.9 mm). ESC3 has no extra cap (C30 deleted, D77). VDD abs max 5.5 V vs 4.35 V cell + ripple + regen 4.7–5.25 V (V3c). Ramp: DS Table 4.7 RAMPVDD >= 10 us/V vs about 1 us/V at a hot plug (spec 4.1): Matrix-parity deviation, V3c 20 hot plugs; a 2.2–4.7 Ohm VDD R (tau 0.5 us) does not fix it.
DShot (P4-8): R12 2.4k at RTX. At a 2.5 V cell the 3.3 V DShot high sits up to 0.53 V over P0.5's VDD + 0.3 V abs max; R12 holds the clamp current to <= 0.35 mA ((3.33 − 2.5) V / 2.4k; about 0.1–0.2 mA after the clamp drop) against 50 mA per pin. RSTb/C2CK: internal pull-up only (PINMAP F12).
C2D / C2CK: TP1 / TP2, filled test vias (D67), bench recovery only: NextPCB pre-programs Bluejay; updates via Betaflight passthrough.

BEMF network: 10k series + 1k neutral star

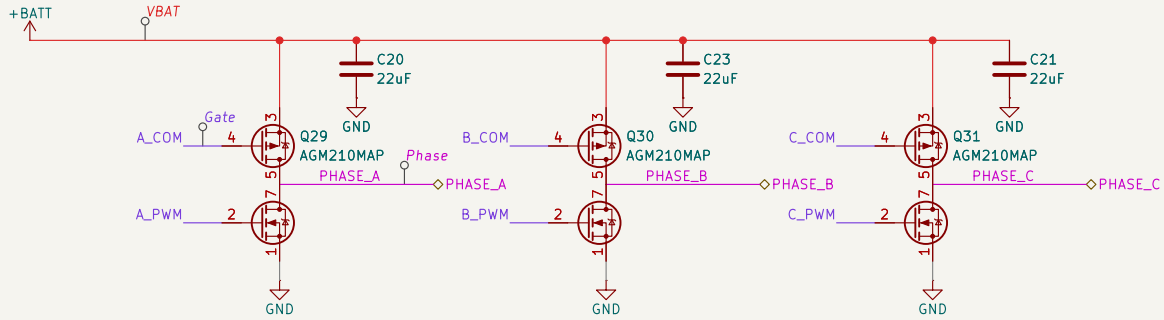


Source: tinyPEPPER single_esc.sch (CERN-OHL v1.2, wire trace in research 08): 10k series from each phase to its comparator input (BEMF_A P0.3, BEMF_B P0.1, BEMF_C P0.2) and 1k from each phase to the virtual neutral (P0.4 V_Mux, CMP0 negative). No leg to GND: at 1S the phase stays within the cell + one body-diode drop, and the 10k limit the input clamp current to < 0.1 mA. 1k star, worst case one phase low and two high: 2/3 x 5.25 V / 1k = 3.5 mA, 12 mW per 0201 (50 mW part). P4-8: the netlist puts BEMF_A/B/C and NEUTRAL on U6 pins 19/1/20 and 18 = A.inc A_Mux 3 / B_Mux 1 / C_Mux 2 / V_Mux 4.

ESC channel, one file placed 4x. The references drawn are ESC1's; the root sheet offsets them per instance (ESC2 U+1 Q+3 R+7 C+6 TP+2, ESC3 x2, ESC4 x3), matching bom_plan.json:
ESC1 M1 rear right: U6 C19, hot loop C20 C23 C21, R12–R18, Q29–Q31, TP1/TP2
ESC2 M2 front right: U7 C25, hot loop C26 C29 C27, R19–R25, Q32–Q34, TP3/TP4
ESC3 M3 rear left: U8 C31, hot loop C32 C35 C33, R26–R32, Q35–Q37, TP5/TP6
ESC4 M4 front left: U9 C37, hot loop C38 C41 C39, R33–R39, Q38–Q40, TP7/TP8
Pins: DSHOT (left, root net /MOTORn: FC GPIO4/6/18/20); PHASE_A/B/C (right, local label + hierarchical label, net /ESCn/PHASE_x, motor lands J4/J7/J10/J13). In-sheet nets: x_COM, x_PWM (Gate class), BEMF_x, NEUTRAL, RTX, C2D, C2CK. Netclass flags: Phase (PHASE_A), Gate (A_COM), VBAT (+BATT rail); setup_board.py patterns /ESC?/PHASE_?, /ESC?/?_COM, /ESC?/?_PWM assign the rest.
Source: copied from OpenAIO hardware/esc_channel.kicad_sch (frames, block titles, C2 test points at the MCU pins, three-column BEMF block, power stage along the bottom) and adapted: AT32F421 + NSG2065Q + 6x DOY180N03T become EFM8BB51 + 3x AGM210MAP (tinyPEPPER direct drive, Matrix 1S 5IN1 II stage, D56), the 3S divider becomes the tinyPEPPER 1S star.

Power-up (P4-11, PINMAP F16): battery first, the EFM8 runs from the cell within 0.4 ms (POR 1.76 V, tPOR <= 390 us) and the Bluejay bootloader turns on the 70–220k weak pull-ups (RTX included) while the RP2354A IOVDD still ramps (boost soft start + LDO, about 1–2 ms). Worst-case FC pad voltage in that window: the cell, up to 4.35 V (LiHV full), through >= 72k (pull-up + R12), i.e. <= 60 uA: 0.72 V over the 3.63 V unpowered-FT limit, current-limited. Accepted on Matrix parity (G473 FT pads about 4 V unpowered); V9 scopes the pad at power-up; fallback a 3.3 V-domain clamp or a DShot-path enable (not in rev1).
USB only: the PIO DShot idle-high back-feeds +BATT through the EFM8 I/O diode, <= (3.3 V − VF) / 2.4k, about 1 mA per channel; the EFM8s may cycle near POR, +5V stays on USB via the TPS2116 (10k fallback if V9 shows brown-out cycling).

Half-bridges A/B/C: AGM210MAP P+N, one package per phase



AGM210MAP (AGMSEMI, DS VER2.73 p1 top view; D23 rule, P3 symbol/footprint check): 1 S1 N source = GND, 2 G1 N gate = x_PWM, 3 S2 P source = +BATT, 4 G2 P gate = x_COM, 5/6 + EP D2 (P drain) and 7/8 + EP D1 (N drain) = PHASE_x; the two drain pads are joined only by the phase island on the board (never split it). Gates straight from the EFM8: no gate resistors and no OR options (tinyPEPPER / Matrix II field record; bom_plan v3 has none); the about 0.15 A gate-edge current for tens of ns is a transient above the EFM8's 50 mA per-pin abs max, accepted as on the Matrix II (spec 4.4). VGS +-12 V, VDS 20 V abs max vs VDD <= 5.25 V regen peak. Hot loop (D77, D108): one cap per AGM210MAP, C20 / C23 / C21 22 uF 6.3 V X6S 0603 (Murata GRM188C80J226ME15D, about 7–10 uF at 4.2 V) from pin 3 S2 (+BATT) to pin 1 S1 (GND) at the cell's lead side, microvias in both pads, so the commutation loop closes at the pins (Matrix II: one cap per FET); the three are also the ESC +BATT bulk. Bluejay: first power-up A_X_10_96 (about 200 ns; DS td(off) + tf = 58 ns N at RGEN 3.3 Ohm, 84 ns P at 1 Ohm, longer on EFM8 port drive), then A_X_5_96 (Matrix II) after the V2 shoot-through check (PINMAP 7.3); the DS Qg of 23 / 33 nC disagrees 3–5x with Ciss 950 pF, so V2 measures the gate edges.

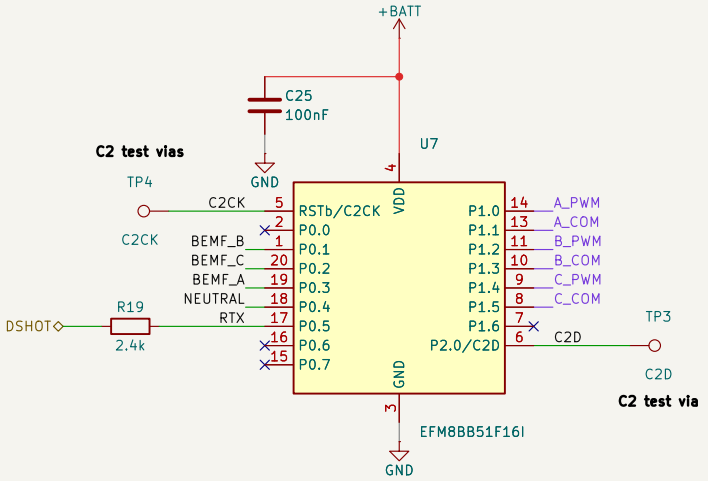
OpenDrone

Sheet: /ESC1/
File: esc_channel.kicad_sch

Title: ESC channel: EFM8BB51 + 3x AGM210MAP

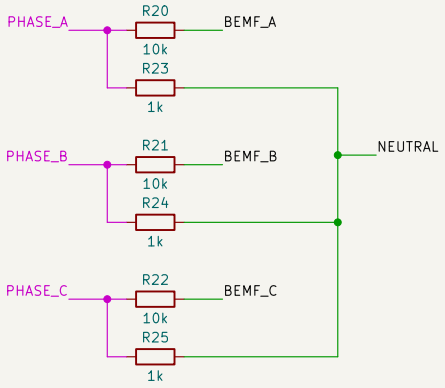
Size: A4	Date: 2026-10-09	Rev: rev1
KiCad E.D.A. 10.0.6		Id: 3/14

ESC MCU: EFM8BB51, Bluejay BB51 layout A



Pin use = Bluejay src/Layouts/BB51/A.inc (0368d11): P0.1 B_Mux, P0.2 C_Mux, P0.3 A_Mux, P0.4 V_Mux (neutral), P0.5 RTX; P1.0/P1.1 A_Pwm/A_Com, P1.2/P1.3 B, P1.4/P1.5 C (PWM active high = N gate, COM active low = P gate, all push-pull); P2.0 DebugPin = C2D. Pin numbers: EFM8BB51 data sheet Rev 1.0 Table 6.1. P0.0, P0.6, P0.7, P1.6 unused: no-connect (PINMAP 7.1).
VDD pin 4 straight on +BATT, no series R (it must track the P sources); C19 100 nF X6S at pin 4 / GND pin 3. The data sheet's 1 uF (Fig. 5.1) is the phase B hot-loop cap C23 22 uF X6S 0603 (D108; about 7–10 uF effective at 4.2 V); BR-05 as amended (P4 critique): within 4 mm of pin 4 through the +BATT plane, 100 nF at the pin (v3 cell site 3.4–3.9 mm). ESC3 has no extra cap (C30 deleted, D77). VDD abs max 5.5 V vs 4.35 V cell + ripple + regen 4.7–5.25 V (V3c). Ramp: DS Table 4.7 RAMPVDD >= 10 us/V vs about 1 us/V at a hot plug (spec 4.1): Matrix-parity deviation, V3c 20 hot plugs; a 2.2–4.7 Ohm VDD R (tau 0.5 us) does not fix it.
DShot (P4-8): R12 2.4k at RTX. At a 2.5 V cell the 3.3 V DShot high sits up to 0.53 V over P0.5's VDD + 0.3 V abs max; R12 holds the clamp current to <= 0.35 mA ((3.33 − 2.5) V / 2.4k; about 0.1–0.2 mA after the clamp drop) against 50 mA per pin. RSTb/C2CK: internal pull-up only (PINMAP F12).
C2D / C2CK: TP1 / TP2, filled test vias (D67), bench recovery only: NextPCB pre-programs Bluejay; updates via Betaflight passthrough.

BEMF network: 10k series + 1k neutral star

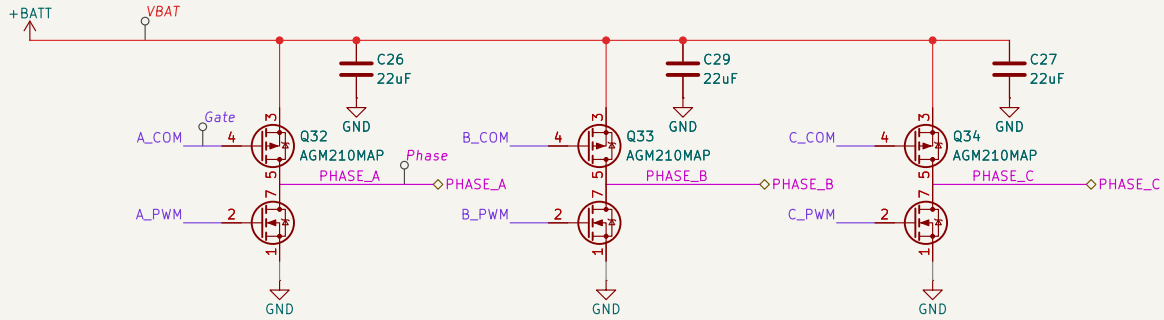


Source: tinyPEPPER single_esc.sch (CERN-OHL v1.2, wire trace in research 08): 10k series from each phase to its comparator input (BEMF_A P0.3, BEMF_B P0.1, BEMF_C P0.2) and 1k from each phase to the virtual neutral (P0.4 V_Mux, CMP0 negative). No leg to GND: at 1S the phase stays within the cell + one body-diode drop, and the 10k limit the input clamp current to < 0.1 mA. 1k star, worst case one phase low and two high: 2/3 x 5.25 V / 1k = 3.5 mA, 12 mW per 0201 (50 mW part). P4-8: the netlist puts BEMF_A/B/C and NEUTRAL on U6 pins 19/1/20 and 18 = A.inc A_Mux 3 / B_Mux 1 / C_Mux 2 / V_Mux 4.

ESC channel, one file placed 4x. The references drawn are ESC1's; the root sheet offsets them per instance (ESC2 U+1 Q+3 R+7 C+6 TP+2, ESC3 x2, ESC4 x3), matching bom_plan.json:
ESC1 M1 rear right: U6 C19, hot loop C20 C23 C21, R12-R18, Q29-Q31, TP1/TP2
ESC2 M2 front right: U7 C25, hot loop C26 C29 C27, R19-R25, Q32-Q34, TP3/TP4
ESC3 M3 rear left: U8 C31, hot loop C32 C35 C33, R26-R32, Q35-Q37, TP5/TP6
ESC4 M4 front left: U9 C37, hot loop C38 C41 C39, R33-R39, Q38-Q40, TP7/TP8
Pins: DSHOT (left, root net /MOTORn: FC GPIO4/6/18/20); PHASE_A/B/C (right, local label + hierarchical label, net /ESCn/PHASE_x, motor lands J4/J7/J10/J13). In-sheet nets: x_COM, x_PWM (Gate class), BEMF_x, NEUTRAL, RTX, C2D, C2CK. Netclass flags: Phase (PHASE_A), Gate (A_COM), VBAT (+BATT rail); setup_board.py patterns /ESC?/PHASE_?, /ESC?/?_COM, /ESC?/?_PWM assign the rest.
Source: copied from OpenAIO hardware/esc_channel.kicad_sch (frames, block titles, C2 test points at the MCU pins, three-column BEMF block, power stage along the bottom) and adapted: AT32F421 + NSG2065Q + 6x DOY180N03T become EFM8BB51 + 3x AGM210MAP (tinyPEPPER direct drive, Matrix 1S 5IN1 II stage, D56), the 3S divider becomes the tinyPEPPER 1S star.

Power-up (P4-11, PINMAP F16): battery first, the EFM8 runs from the cell within 0.4 ms (POR 1.76 V, tPOR <= 390 us) and the Bluejay bootloader turns on the 70–220k weak pull-ups (RTX included) while the RP2354A IOVDD still ramps (boost soft start + LDO, about 1–2 ms). Worst-case FC pad voltage in that window: the cell, up to 4.35 V (LiHV full), through >= 72k (pull-up + R12), i.e. <= 60 uA: 0.72 V over the 3.63 V unpowered-FT limit, current-limited. Accepted on Matrix parity (G473 FT pads about 4 V unpowered); V9 scopes the pad at power-up; fallback a 3.3 V-domain clamp or a DShot-path enable (not in rev1).
USB only: the PIO DShot idle-high back-feeds +BATT through the EFM8 I/O diode, <= (3.3 V − VF) / 2.4k, about 1 mA per channel; the EFM8s may cycle near POR, +5V stays on USB via the TPS2116 (10k fallback if V9 shows brown-out cycling).

Half-bridges A/B/C: AGM210MAP P+N, one package per phase



AGM210MAP (AGMSEMI, DS VER2.73 p1 top view; D23 rule, P3 symbol/footprint check): 1 S1 N source = GND, 2 G1 N gate = x_PWM, 3 S2 P source = +BATT, 4 G2 P gate = x_COM, 5/6 + EP D2 (P drain) and 7/8 + EP D1 (N drain) = PHASE_x; the two drain pads are joined only by the phase island on the board (never split it). Gates straight from the EFM8: no gate resistors and no 0R options (tinyPEPPER / Matrix II field record; bom_plan v3 has none); the about 0.15 A gate-edge current for tens of ns is a transient above the EFM8's 50 mA per-pin abs max, accepted as on the Matrix II (spec 4.4). VGS +-12 V, VDS 20 V abs max vs VDD <= 5.25 V regen peak. Hot loop (D77, D108): one cap per AGM210MAP, C20 / C23 / C21 22 uF 6.3 V X6S 0603 (Murata GRM188C80J226ME15D, about 7–10 uF at 4.2 V) from pin 3 S2 (+BATT) to pin 1 S1 (GND) at the cell's lead side, microvias in both pads, so the commutation loop closes at the pins (Matrix II: one cap per FET); the three are also the ESC +BATT bulk. Bluejay: first power-up A_X_10_96 (about 200 ns; DS td(off) + tf = 58 ns N at RGEN 3.3 Ohm, 84 ns P at 1 Ohm, longer on EFM8 port drive), then A_X_5_96 (Matrix II) after the V2 shoot-through check (PINMAP 7.3); the DS Qg of 23 / 33 nC disagrees 3-5x with Ciss 950 pF, so V2 measures the gate edges.

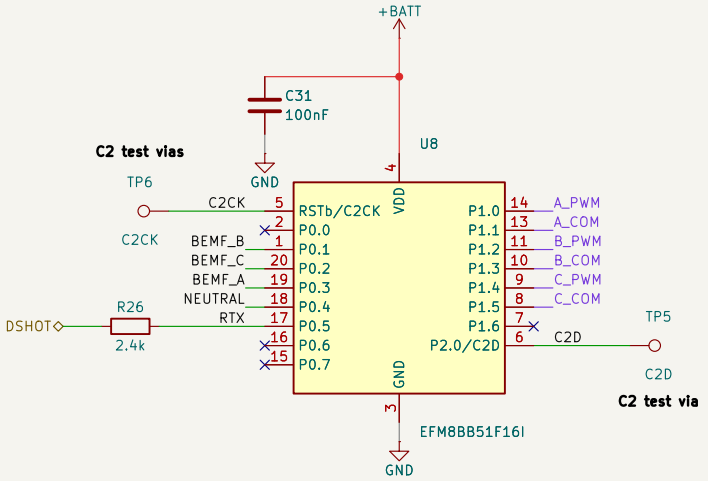
OpenDrone

Sheet: /ESC2/
File: esc_channel.kicad_sch

Title: ESC channel: EFM8BB51 + 3x AGM210MAP

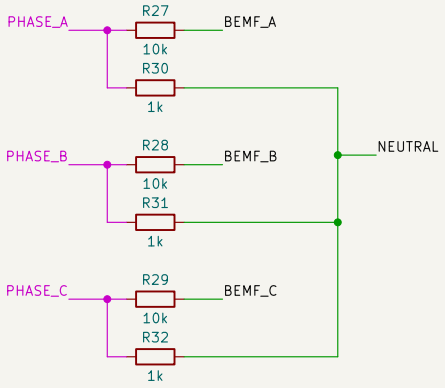
Size: A4	Date: 2026-10-09	Rev: rev1
KiCad E.D.A. 10.0.6		Id: 4/14

ESC MCU: EFM8BB51, Bluejay BB51 layout A



Pin use = Bluejay src/Layouts/BB51/A.inc (0368d11): P0.1 B_Mux, P0.2 C_Mux, P0.3 A_Mux, P0.4 V_Mux (neutral), P0.5 RTX; P1.0/P1.1 A_Pwm/A_Com, P1.2/P1.3 B, P1.4/P1.5 C (PWM active high = N gate, COM active low = P gate, all push-pull); P2.0 DebugPin = C2D. Pin numbers: EFM8BB51 data sheet Rev 1.0 Table 6.1. P0.0, P0.6, P0.7, P1.6 unused: no-connect (PINMAP 7.1).
VDD pin 4 straight on +BATT, no series R (it must track the P sources); C19 100 nF X6S at pin 4 / GND pin 3. The data sheet's 1 uF (Fig. 5.1) is the phase B hot-loop cap C23 22 uF X6S 0603 (D108; about 7–10 uF effective at 4.2 V); BR-05 as amended (P4 critique): within 4 mm of pin 4 through the +BATT plane, 100 nF at the pin (v3 cell site 3.4–3.9 mm). ESC3 has no extra cap (C30 deleted, D77). VDD abs max 5.5 V vs 4.35 V cell + ripple + regen 4.7–5.25 V (V3c). Ramp: DS Table 4.7 RAMPVDD >= 10 us/V vs about 1 us/V at a hot plug (spec 4.1): Matrix-parity deviation, V3c 20 hot plugs; a 2.2–4.7 Ohm VDD R (tau 0.5 us) does not fix it.
DShot (P4-8): R12 2.4k at RTX. At a 2.5 V cell the 3.3 V DShot high sits up to 0.53 V over P0.5's VDD + 0.3 V abs max; R12 holds the clamp current to <= 0.35 mA ((3.33 − 2.5) V / 2.4k; about 0.1–0.2 mA after the clamp drop) against 50 mA per pin. RSTb/C2CK: internal pull-up only (PINMAP F12).
C2D / C2CK: TP1 / TP2, filled test vias (D67), bench recovery only: NextPCB pre-programs Bluejay; updates via Betaflight passthrough.

BEMF network: 10k series + 1k neutral star

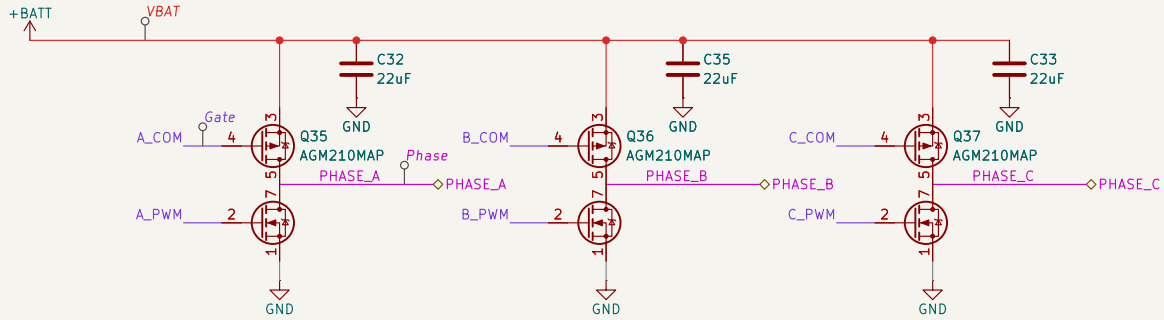


Source: tinyPEPPER single_esc.sch (CERN-OHL v1.2, wire trace in research 08): 10k series from each phase to its comparator input (BEMF_A P0.3, BEMF_B P0.1, BEMF_C P0.2) and 1k from each phase to the virtual neutral (P0.4 V_Mux, CMP0 negative). No leg to GND: at 1S the phase stays within the cell + one body-diode drop, and the 10k limit the input clamp current to < 0.1 mA. 1k star, worst case one phase low and two high: 2/3 x 5.25 V / 1k = 3.5 mA, 12 mW per 0201 (50 mW part). P4-8: the netlist puts BEMF_A/B/C and NEUTRAL on U6 pins 19/1/20 and 18 = A.inc A_Mux 3 / B_Mux 1 / C_Mux 2 / V_Mux 4.

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Pins: DSHOT (left, root net /MOTORn: FC GPIO4/6/18/20); PHASE_A/B/C (right, local label + hierarchical label, net /ESCn/PHASE_x, motor lands J4/J7/J10/J13). In-sheet nets: x_COM, x_PWM (Gate class), BEMF_x, NEUTRAL, RTX, C2D, C2CK. Netclass flags: Phase (PHASE_A), Gate (A_COM), VBAT (+BATT rail); setup_board.py patterns /ESC?/PHASE_?, /ESC?/?_COM, /ESC?/?_PWM assign the rest.
Source: copied from OpenAIO hardware/esc_channel.kicad_sch (frames, block titles, C2 test points at the MCU pins, three-column BEMF block, power stage along the bottom) and adapted: AT32F421 + NSG2065Q + 6x DOY180N03T become EFM8BB51 + 3x AGM210MAP (tinyPEPPER direct drive, Matrix 1S 5IN1 II stage, D56), the 3S divider becomes the tinyPEPPER 1S star.

Power-up (P4-11, PINMAP F16): battery first, the EFM8 runs from the cell within 0.4 ms (POR 1.76 V, tPOR <= 390 us) and the Bluejay bootloader turns on the 70–220k weak pull-ups (RTX included) while the RP2354A IOVDD still ramps (boost soft start + LDO, about 1–2 ms). Worst-case FC pad voltage in that window: the cell, up to 4.35 V (LiHV full), through >= 72k (pull-up + R12), i.e. <= 60 uA: 0.72 V over the 3.63 V unpowered-FT limit, current-limited. Accepted on Matrix parity (G473 FT pads about 4 V unpowered); V9 scopes the pad at power-up; fallback a 3.3 V-domain clamp or a DShot-path enable (not in rev1).
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Half-bridges A/B/C: AGM210MAP P+N, one package per phase



AGM210MAP (AGMSEMI, DS VER2.73 p1 top view; D23 rule, P3 symbol/footprint check): 1 S1 N source = GND, 2 G1 N gate = x_PWM, 3 S2 P source = +BATT, 4 G2 P gate = x_COM, 5/6 + EP D2 (P drain) and 7/8 + EP D1 (N drain) = PHASE_x; the two drain pads are joined only by the phase island on the board (never split it). Gates straight from the EFM8: no gate resistors and no 0R options (tinyPEPPER / Matrix II field record; bom_plan v3 has none); the about 0.15 A gate-edge current for tens of ns is a transient above the EFM8's 50 mA per-pin abs max, accepted as on the Matrix II (spec 4.4). VGS +-12 V, VDS 20 V abs max vs VDD <= 5.25 V regen peak. Hot loop (D77, D108): one cap per AGM210MAP, C20 / C23 / C21 22 uF 6.3 V X6S 0603 (Murata GRM188C80J226ME15D, about 7–10 uF at 4.2 V) from pin 3 S2 (+BATT) to pin 1 S1 (GND) at the cell's lead side, microvias in both pads, so the commutation loop closes at the pins (Matrix II: one cap per FET); the three are also the ESC +BATT bulk. Bluejay: first power-up A_X_10_96 (about 200 ns; DS td(off) + tf = 58 ns N at RGEN 3.3 Ohm, 84 ns P at 1 Ohm, longer on EFM8 port drive), then A_X_5_96 (Matrix II) after the V2 shoot-through check (PINMAP 7.3); the DS Qg of 23 / 33 nC disagrees 3–5x with Ciss 950 pF, so V2 measures the gate edges.

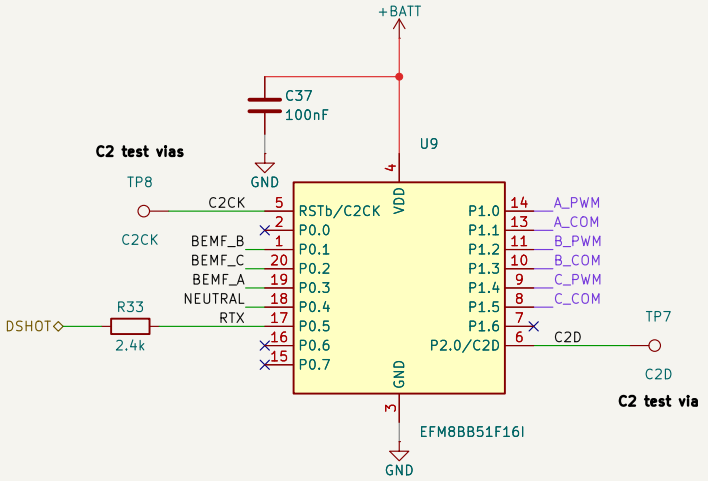
OpenDrone

Sheet: /ESC3/
File: esc_channel.kicad_sch

Title: ESC channel: EFM8BB51 + 3x AGM210MAP

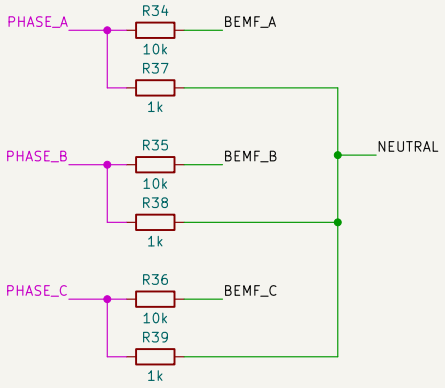
Size: A4	Date: 2026-10-09	Rev: rev1
KiCad E.D.A. 10.0.6		Id: 5/14

ESC MCU: EFM8BB51, Bluejay BB51 layout A



Pin use = Bluejay src/Layouts/BB51/A.inc (0368d11): P0.1 B_Mux, P0.2 C_Mux, P0.3 A_Mux, P0.4 V_Mux (neutral), P0.5 RTX; P1.0/P1.1 A_Pwm/A_Com, P1.2/P1.3 B, P1.4/P1.5 C (PWM active high = N gate, COM active low = P gate, all push-pull); P2.0 DebugPin = C2D. Pin numbers: EFM8BB51 data sheet Rev 1.0 Table 6.1. P0.0, P0.6, P0.7, P1.6 unused: no-connect (PINMAP 7.1).
VDD pin 4 straight on +BATT, no series R (it must track the P sources); C19 100 nF X6S at pin 4 / GND pin 3. The data sheet's 1 uF (Fig. 5.1) is the phase B hot-loop cap C23 22 uF X6S 0603 (D108; about 7–10 uF effective at 4.2 V); BR-05 as amended (P4 critique): within 4 mm of pin 4 through the +BATT plane, 100 nF at the pin (v3 cell site 3.4–3.9 mm). ESC3 has no extra cap (C30 deleted, D77). VDD abs max 5.5 V vs 4.35 V cell + ripple + regen 4.7–5.25 V (V3c). Ramp: DS Table 4.7 RAMPVDD >= 10 us/V vs about 1 us/V at a hot plug (spec 4.1): Matrix-parity deviation, V3c 20 hot plugs; a 2.2–4.7 Ohm VDD R (tau 0.5 us) does not fix it.
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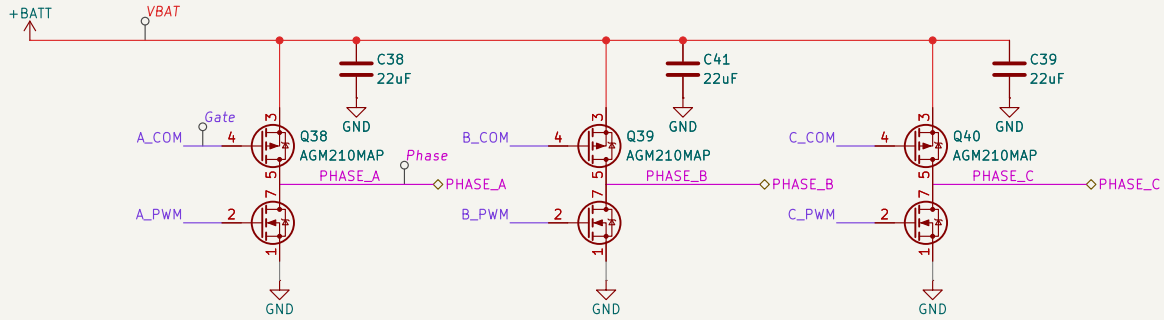


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ESC channel, one file placed 4x. The references drawn are ESC1's; the root sheet offsets them per instance (ESC2 U+1 Q+3 R+7 C+6 TP+2, ESC3 x2, ESC4 x3), matching bom_plan.json:
ESC1 M1 rear right: U6 C19, hot loop C20 C23 C21, R12-R18, Q29-Q31, TP1/TP2
ESC2 M2 front right: U7 C25, hot loop C26 C29 C27, R19-R25, Q32-Q34, TP3/TP4
ESC3 M3 rear left: U8 C31, hot loop C32 C35 C33, R26-R32, Q35-Q37, TP5/TP6
ESC4 M4 front left: U9 C37, hot loop C38 C41 C39, R33-R39, Q38-Q40, TP7/TP8
Pins: DSHOT (left, root net /MOTORn: FC GPIO4/6/18/20); PHASE_A/B/C (right, local label + hierarchical label, net /ESCn/PHASE_x, motor lands J4/J7/J10/J13). In-sheet nets: x_COM, x_PWM (Gate class), BEMF_x, NEUTRAL, RTX, C2D, C2CK. Netclass flags: Phase (PHASE_A), Gate (A_COM), VBAT (+BATT rail); setup_board.py patterns /ESC?/PHASE_?, /ESC?/?_COM, /ESC?/?_PWM assign the rest.
Source: copied from OpenAIO hardware/esc_channel.kicad_sch (frames, block titles, C2 test points at the MCU pins, three-column BEMF block, power stage along the bottom) and adapted: AT32F421 + NSG2065Q + 6x DOY180N03T become EFM8BB51 + 3x AGM210MAP (tinyPEPPER direct drive, Matrix 1S 5IN1 II stage, D56), the 3S divider becomes the tinyPEPPER 1S star.

Power-up (P4-11, PINMAP F16): battery first, the EFM8 runs from the cell within 0.4 ms (POR 1.76 V, tPOR <= 390 us) and the Bluejay bootloader turns on the 70–220k weak pull-ups (RTX included) while the RP2354A IOVDD still ramps (boost soft start + LDO, about 1–2 ms). Worst-case FC pad voltage in that window: the cell, up to 4.35 V (LiHV full), through >= 72k (pull-up + R12), i.e. <= 60 uA: 0.72 V over the 3.63 V unpowered-FT limit, current-limited. Accepted on Matrix parity (G473 FT pads about 4 V unpowered); V9 scopes the pad at power-up; fallback a 3.3 V-domain clamp or a DShot-path enable (not in rev1).
USB only: the PIO DShot idle-high back-feeds +BATT through the EFM8 I/O diode, <= (3.3 V − VF) / 2.4k, about 1 mA per channel; the EFM8s may cycle near POR, +5V stays on USB via the TPS2116 (10k fallback if V9 shows brown-out cycling).

Half-bridges A/B/C: AGM210MAP P+N, one package per phase



AGM210MAP (AGMSEMI, DS VER2.73 p1 top view; D23 rule, P3 symbol/footprint check): 1 S1 N source = GND, 2 G1 N gate = x_PWM, 3 S2 P source = +BATT, 4 G2 P gate = x_COM, 5/6 + EP D2 (P drain) and 7/8 + EP D1 (N drain) = PHASE_x; the two drain pads are joined only by the phase island on the board (never split it). Gates straight from the EFM8: no gate resistors and no OR options (tinyPEPPER / Matrix II field record; bom_plan v3 has none); the about 0.15 A gate-edge current for tens of ns is a transient above the EFM8's 50 mA per-pin abs max, accepted as on the Matrix II (spec 4.4). VGS +-12 V, VDS 20 V abs max vs VDD <= 5.25 V regen peak. Hot loop (D77, D108): one cap per AGM210MAP, C20 / C23 / C21 22 uF 6.3 V X6S 0603 (Murata GRM188C80J226ME15D, about 7–10 uF at 4.2 V) from pin 3 S2 (+BATT) to pin 1 S1 (GND) at the cell's lead side, microvias in both pads, so the commutation loop closes at the pins (Matrix II: one cap per FET); the three are also the ESC +BATT bulk. Bluejay: first power-up A_X_10_96 (about 200 ns; DS td(off) + tf = 58 ns N at RGEN 3.3 Ohm, 84 ns P at 1 Ohm, longer on EFM8 port drive), then A_X_5_96 (Matrix II) after the V2 shoot-through check (PINMAP 7.3); the DS Qg of 23 / 33 nC disagrees 3–5x with Ciss 950 pF, so V2 measures the gate edges.

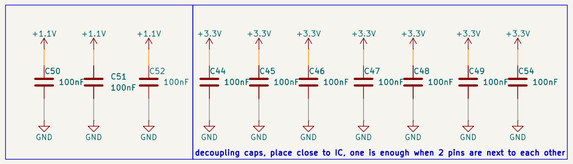
OpenDrone

Sheet: /ESC4/
File: esc_channel.kicad_sch

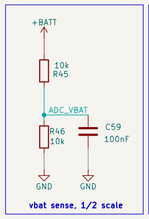
Title: ESC channel: EFM8BB51 + 3x AGM210MAP

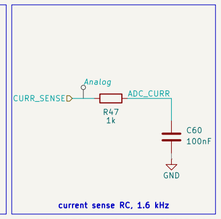
Size: A4	Date: 2026-10-09	Rev: rev1
KiCad E.D.A. 10.0.6		Id: 6/14

RP2354A: Smaller variant with integrated flash (2MB)



DVDD 6/23/39: C50–C52, IOVDD 1/11/20/30/38: C44–C48, one per pin. C49 sits at ADC_AVDD pin 44 and also serves IOVDD 45 (BR-08; RP2350 DS 6.1.5: 100 nF close to ADC_AVDD). C54 <= 1 mm from QSPI_IOVDD pin 54 (the in-package flash supply, 2.97–3.63 V) and also serves USB_OTP_VDD 53 (BR-08, P4-17 placement condition). All Murata GRM033C81E104KE14D 100 nF X6S 0201 (BR-18). Copied from OpenFC-Lite-Mini rp2350a (7 caps for the 9 3V3 pins there too); its 4.7 uF on this row moved to the SMPS block (C56).

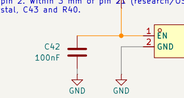




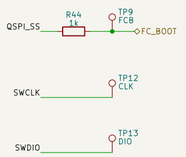
VBAT: R45/R46 10k/10k (house 100k/10k). 4.35 V -> 2.18 V on GPIO27 (ADC1, not 5 V tolerant). vbat_scale 200/100: 5 kOhm with C59 = 318 Hz. CURR: R47 1k + C60 100 nF = 1.6 kHz on GPIO26 (ADC0); INA186A3 on POWER, 50 mV/A, ibata_scale 500. Copied from OpenFC-Lite-Mini rp2350a (vbat + ESC current).

12 MHz CMOS oscillator on XIN, XOUT open (D100)

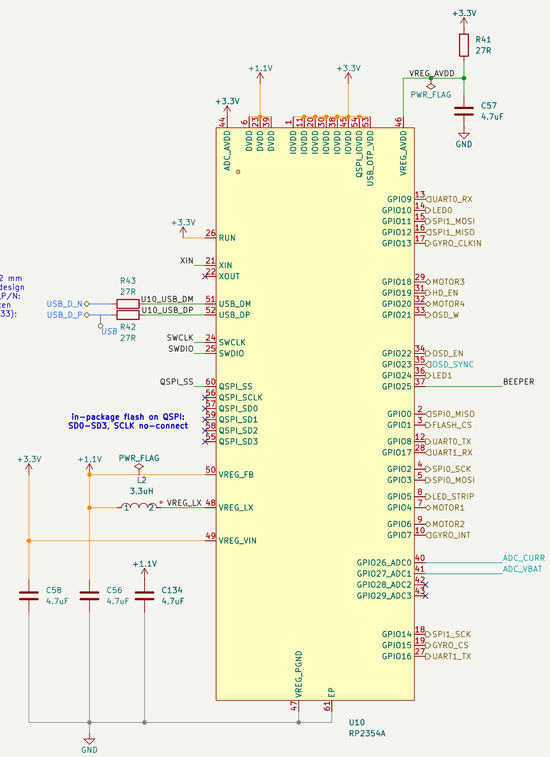
X1 YXC OT201612MJBA4SL 12 MHz CMOS oscillator (2016) drives XIN (pin 21) as a single-ended clock; XOUT (pin 22) stays open, no trace (RP2350 DS 8.2 / 14.9: square wave on XIN only, XOUT floating; BOOTSEL defaults to 12 MHz). EN tied to +3.3V. C42 100 nF at VDD pin 4, GND pad next to pin 2. Within 3 mm of pin 21 (research/OSCILLATOR-03.md). D99(a)/D100 replace the D78 crystal, C43 and R40.



Boot strap + SWD test vias

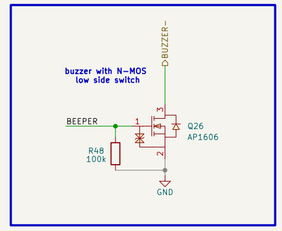


R44 1k from QSPI_SS (pin 60, BOOT) to FC_BOOT: the shared boot button SW1 pulls FC_BOOT low through D8 (RX sheet, D65): low at reset = BOOTSEL / UF2. 1k (RPi hardware guide R6): CSn <= 0.51 V with the diode drop (VIL 0.8 V). Keep the QSPI_SS-to-R44 stub <= 1 mm (in-package flash CS. RPi guide R6): any length after R44. TP9 FCB, TP12 CLK (SWCLK), TP13 DIO (SWDIO): mask-open filled 0.35/0.20 vias (D67), no silk, out of the BOM and the CPL, listed in the pinout doc (the house sheet leaves SWD open). Source: OpenFC-Lite-Mini rp2350a BOOTSEL strap, button moved to the RX sheet.



R42/R43 27R within 1–2 mm of pins 52/51 (RPi HW design 5.1, house 30R). USB_D_P/N: 90 ohm pair over unbroken GND, length-matched (D33): D7 ESD + J31 on PADS. Pin-side nets U10_USB_DP/DM.

RP2350 1.1V DC–DC SMPS: follow the datasheet placement. RP2350 DS 6.3.8 (binding): C58 = CIN at VREG_VIN, C56 = COUT on +1.1V with VREG_FB taken from the C56 pad (not under L2); C58, L2 and C56 on the U10 side, smallest hot loops: CIN/COUT GND to VREG_PGND at one point with 2 vias; no copper under L2 / VREG_LX on the next layer. L2 AOTA-B201610S3R3 is polarity-marked: orientation as DS Fig. 26 / 28 (pin 2 = VREG_LX as on the flown house board). P5 condition (tools/check_p5_conditions.py SMPS): C58, L2, C56 and C57 on U10's side within 3 mm of pins 49 / 48 / 50 / 46. R41 27R + C57 4.7 uF filter VREG_AVDD (1.25 kHz, BR-21; 3.135–3.63 V): C57 gets its own GND via to the EP. DS: CIN >= 4.7 uF. COUT 4.7 uF +/-20 %: C134 = the 2nd 4.7 uF on +1.1V that DS 6.3.8 recommends at DVDD pin 23 (fitted per the 2026-10-09 ruling, house board fits two): place it at pin 23.



Q26 AP1606 (copied from OpenFC-Lite-Mini rp2350a, Q5 there): 20 V, 0.7 A; Vgs(th) 0.4–1.2 V; RDS(on) <= 220 mOhm at Vgs 2.5 V (4.1 ohm at 1.8 V). R48 100k pulldown (BR-08 P4-11 "pulldown on buzzer, no PWM"): BEEPER low through reset; BZ+ = +5V and BZ- = DBUZZER- pad row (PADS).

GPIO map (D111, placement V4_B4): GPIO0 QSPIO_MISO / GPIO1 FLASH_CS / GPIO2 QSPIO_SCK / GPIO3 QSPIO_MOSI (FLASH_CS GPIO1 MOSI GPIO3 SCK GPIO2 MISO, BLACKBOX). SPI1 gyro GPIO11 MOSI GPIO12 MISO GPIO13 SCK GPIO9 CS (GYRO_CLKIN GPIO13 (ICM-42688-P only). CRSF to the ESP32 ELRS on hardware UART1 GPIO8 TX / 9 RX (nets UART0_TX/RX keep their names). TX1/RX1 pads (DJI O4 Lite MSP DisplayPort) on hardware UART0 GPIO16 / 17 (nets UART1_TX/RX). PIO0 bidirectional DShot MOTOR1–4 on GPIO4 / 6 / 18 / 20 (2.4k at the EFM8). LED_STRIP GPIO5 (direct, F1). LED0 GPIO10. LED1 GPIO24 (LED sheet, sinking). HD_EN GPIO19 (PINIO1: high = O4 on, analog VTX off). PIO2 framebuffer OSD W / EN / SYNC on the consecutive GPIO21 / 22 / 23 (osd_pico.c). BEEPER GPIO25. ADC_CURR GPIO26 (ADC0). ADC_VBAT GPIO27 (ADC1). GPIO28/29 spare, no-connect.

SHEET RP2350A, source: OpenFC-Lite-Mini/hardware/rp2350a.kicad_sch (flown Rev2), copied block by block with its layout, frames and text style. Changes: USB-C + CC removed (USB = JST SH J31 + D7 on PADS; 27R series R42/R43 stay at U10, D52 BR-21 value); I2C0 pull-ups removed; the inverting WS2812 stage is not copied (F1, BR-04; GPIO5 drives the LED pad); status LEDs moved to the LED sheet; BOOTSEL button replaced by R44 + FC_BOOT (shared button on RX); SWD to test vias (D67); VBAT divider 10k/10k; 12 MHz clock = YXC OT201612MJBA4SL CMOS oscillator on XIN (D100); VREG_AVDD 27R; two 4.7 uF on +1.1V (C56 at VREG_FB. C134 at DVDD 23). GPIO map: research/PINMAP.md 2–3 (28 of 30 GPIOs. GPIO28/29 no-connect; D111 map of placement V4_B4); Betaflight config PINMAP 5.

P4–11 power-up and 5 V-tolerant pads (PINMAP 4.5, F16; RP2350 DS Table 1433): an FT pad takes <= 3.63 V while IOVDD is 0–1.8 V (4.2 V at 2.5 V, 5.5 V at 3.3 V); the ADC pads GPIO26–29 take <= IOVDD + 0.5 V.
– MOTOR1–4 (GPIO4/6/18/20, FT). battery first: each EFM8 weak pull-up (70–220 kOhm to VDD = cell) reaches the pad through its 2.4k before +3.3V is up (boost soft start + LP5912 ramp, about 1–2 ms). Worst case 4.35 V open circuit = 0.72 V over 3.63 V, from >= 72 kOhm (<= 60 uA). Matrix parity (G473 FT pins ~4 V at VDD 0). V9 scopes it: fallback (F16): 3.3 V-domain clamp or DShot-path enable.
– ADC_VBAT (GPIO27): R45/R46 put up to 2.18 V on the pad from 5 kOhm while IOVDD < 1.7 V, so the pad clamp to IOVDD conducts <= 0.3 mA during the ramp. House + Matrix parity: V9 scope.
– HD_EN (GPIO19, FT): R11 2.4k + the BAS16LD from O4_EN (<= 90 uA) hold it at <= 0.22 V before and during reset; inside the pad limits and below the AP1606 HD-gate Vgs(th) min 0.4 V at 25 C (P4-4: about 0.25 V at 85 C, −2.5 mV/K: a hot low-Vth part may hold the analog VTX off until PINIO1 is driven low. R68 47k. V9 hot-soak boot check): driven high it is >= 2.62 V (VOH, 1.4 mA into R11) over Vgs(th) max 1.2 V.
– CRSF UART, SPIO/1, gyro, OSD. CURR_SENSE: every driver is on +3.3V = IOVDD, no excursion. TX1/RX1: the O4 is fed only through U5 (needs HD_EN high), none. USB D+/D−: the host drives only after the device pull-up.
– TX1/RX1 used as the free hardware UART in analog builds (IO-SET row 8: the only free UART since D89): a peripheral on a 5V pad is up before IOVDD: 3.3 V logic only (README): a 5 V TX on RX1 (GPIO17) would exceed 3.63 V at power-up.
– GPIO28/29 (pins 42/43): spare, no-connect (D111 map).

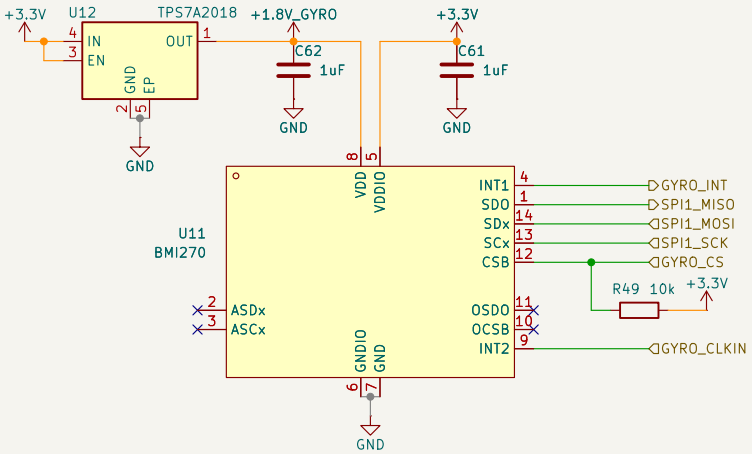
OpenDrone

Sheet: /RP2350A/
File: rp2350a.kicad_sch

Title: RP2354A flight controller MCU

Size: A2	Date: 2026-10-09	**Rev: rev1**
KiCad E.D.A. 10.0.6		Id: 7/14

Universal IMU wiring and LGA-14 land (BMI270, ICM-42688-P)



GYRO BMI270: U11, R49
Source: OpenFC-Lite-Mini imu.kicad_sch (universal LGA-14 land, SPI1 4-wire, R49), copied and adapted. BMI270 on the BOM; TDK ICM-42688-P qualified on the same land (V10 fly-off, one part per revision).
Pins 2/3 (BMI270 ASDx/ASCx, TDK RESV) and 10/11 (OCSB/OSDO, TDK RESV): no-connect. BMI270 DS Table 22: unused ASDx/ASCx go to VDDIO or stay open, never to GND (the house sheet grounded them for the LSM6DSV16X).
Pins 6/7 GND (TDK pin 7 RESV must be GND). VDDIO pin 5 = +3.3V (RP2354A I/O level). VDD pin 8 = +1.8V_GYRO; the BMI270 has no supply-order rule (DS 4.3). Abs max 4.0 V on VDD/VDDIO, I/O up to VDDIO + 0.3 V.
Pin 9: TDK CLKIN = GYRO_CLKIN (RP2354A GPIO6, PWM 32 kHz, driven only with the TDK part); BMI270 INT2, unused, left disabled.
R49 10k: GPIO13 resets with its pull-down on, which would select the chip; R49 holds CSB >= 2.58 V (0.78 x VDDIO with the RP2354A RPD min 36 kOhm at IOVDD 3.3 V, RP2350 DS Table 1683) > VIH 0.7 x VDDIO = 2.31 V.
P5: top side next to U12, LGA rotated 90 deg; GYRO_1_ALIGN from the placed orientation (PINMAP F11).

+1.8V_GYRO GYRO LDO: U12, C61, C62 (D52 BR-09, P4-17)
Source: pcb-agent-commons blocks/LDO_IMU_TPS7A2033 (OpenFC-H7) topology with the 1.8 V part and no ferrite (fed from the +3.3V LDO); the house OpenFC-Lite-Mini also runs the gyro VDD from its own 1.8 V LDO.
TPS7A2018 DQN: 1 OUT, 2 GND, 3 EN, 4 IN, 5 EP to GND. IN = EN = +3.3V: VIN 1.6-6.0 V, VEN 0-6.0 V (abs max 6.5 V), EN never above IN. Load < 1 mA.
No 100 nF at the gyro: C61 = U12 CIN at BMI270 VDDIO pin 5, C62 = U12 COUT at VDD pin 8 (COUT >= 0.47 uF effective, ESR <= 100 mOhm: about 0.6-0.85 uF at 1.8 V). C61 counts in the +3.3V COUT sum (P4-12, POWER).
P5: U12 on the gyro side with OUT facing pin 8; C61 between U12 pin 4 and gyro pin 5, C62 <= 1.5 mm from pin 8 (PS-06), else re-place U12.

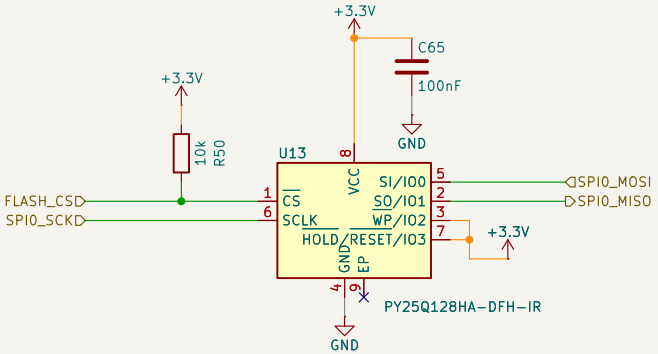
OpenDrone

Sheet: /IMU/
File: imu.kicad_sch

Title: Gyro BMI270 + 1.8 V LDO

Size: A5	Date: 2026-10-09	**Rev: rev1**
KiCad E.D.A. 10.0.6		Id: 8/14

SPI NOR PY25Q128HA: 16 MB blackbox on SPI0 (U13 C65 R50)



Source: OpenFC-Lite-Mini hardware/blackbox.kicad_sch (rev2). Kept: layout, the 10k CS pull-up (R39 -> R50) and the 100n supply cap (C43 -> C65). Changed: its microSD (Card1) became this 16 MB SPI NOR (D66); the SPI0{SCK,MOSI,MISO} bus label became four hierarchical labels (sch_contract.json).

U13 Puya PY25Q128HA-DFH-IR: 128 Mbit, USON-8 3x3 (DF), 2.7-3.6 V, -40..85 C, fC 133 MHz, READ 03h 80 MHz. JEDEC ID 85 20 18 is in the Betaflight m25p16 table: USE_FLASH_M25P16, SPI0 PA18/PA19/PA20, FLASH_CS_PIN PA21 (PINMAP). SPI0 runs at 75 MHz for writes AND reads: Betaflight bus_spi_pico.c spiCalculateDivider() uses an even prescale >= 2 on clk_peri 150 MHz, so the table's 133 / 80 MHz both give 150/2. Writes are inside fC; reads are not guaranteed: tCLQV 6-7 ns vs the 6.67 ns SCK half period plus pad and trace delay (the RP2350 DS gives no SPI receive timing). Gate F7 / V9: erase, write, read back and CRC a full log, status polling under load; fallback = a lower read clock for this JEDEC ID (firmware cap or table entry), upstreamed with the board config.

WP# (3) and HOLD#/RESET# (7) tied to +3.3V: the DS gives an internal pull-up only to the SOP-16 dedicated RESET#, so pin 7 must not float. Delivered SR = 00h, CR = 00h (QE 0, HOLD# function, WPS 0: no block lock) and Betaflight sends 1-bit SPI commands only, so IO2/IO3 never drive against the rail (the ERC pin_to_pin warning on pins 3/7 against a +3.3V power output is this deliberate tie).

R50 10k: CS# must follow VCC at power-up (DS 5.5). RP2354A pads reset with the pull-down on (PDE 1, RPD 36-113 kOhm at 3.3 V): CS >= 36/46 VCC = 0.78 VCC > VIH 0.7 VCC, so U13 stays deselected through reset and boot. U13 and the RP2354A IOVDD share +3.3V: no pin above VCC + 0.5 V.

C65 100 nF X6S at pin 8, GND return straight to pin 4 (ICC max 13 mA read, 12 mA erase; adds 0.1 uF nominal to the +3.3V COUT sum of P4-12). EP (9) = two tie-bar tabs with no electrical function in the DS: no-connect, no vias in the tabs.

Placement (P5): beside the RP2354A on SPI0 (bom_plan region nor, top; either side works). U13, SPI0 and FLASH_CS stay >= 3 mm from the RX antenna hole (spec 4.9 and 11, DRU 'SPI0 3 mm from the RX antenna hole'): SPI0 harmonics fall in the ELRS band, V7 measures desense while logging. C65 at pin 8. Filled+capped 0.35/0.20 via-in-pad allowed in the lead lands, never in the tabs. Betaflight sets SCK to fast slew (bus_spi_pico.c): MOSI and CS keep the reset 4 mA drive and slow slew (EMI lever without parts).
Sourcing: no LCSC listing: NextPCB buys by MPN (D60), the quote confirms it. Same-pinout fallbacks in the Betaflight table (USON-8 4x4 land not staged): PY25Q128HA-QVH-IR, GD25Q128EQIGR.

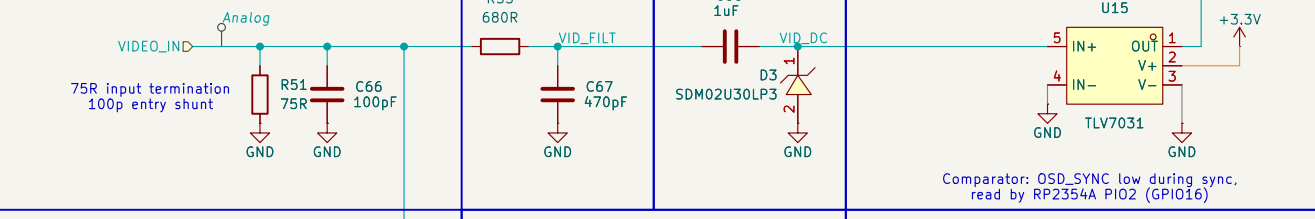
OpenDrone

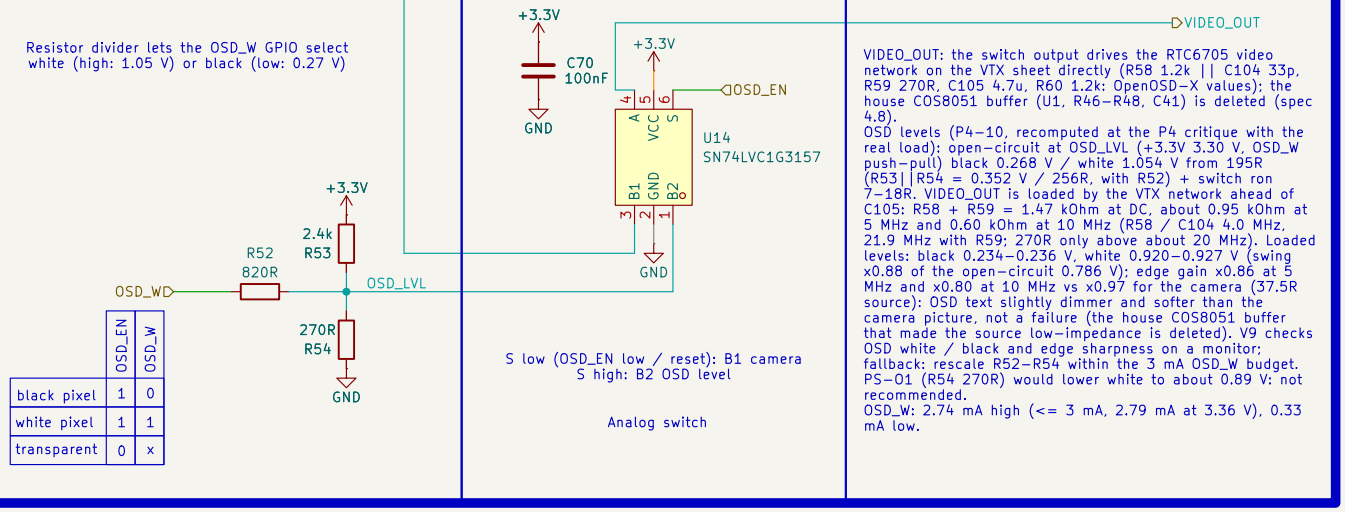
Sheet: /BLACKBOX/
File: blackbox.kicad_sch

Title: Blackbox 16 MB SPI NOR PY25Q128HA

Size: A5	Date: 2026-10-09	Rev: rev1
KiCad E.D.A. 10.0.6		Id: 9/14

VIDEO PATH





Source: OpenFC−Lite−Mini hardware/osd.kicad_sch (rev2, flown), copied and adapted: cells, texts, truth table and part spots kept (A3 −> A4). Same topology: 75R termination, 680R/470p chroma filter, 1u + SDM02U30LP3 sync−tip clamp, TLV7031 comparator, SN74LVC1G3157 camera/level switch (contract source, spec 4.8).
Changed: COS8051 buffer deleted (output = VIDEO_OUT); divider R42 3.9k / R43 12k / R44 1.4k −> R52 820R / R53 2.4k / R54 287R (about 5x down, P4−10); C39 + C42 −> one shared C70 100n X6S (D52 BR−13 + BR−18); C66 100p entry shunt added (spec 11); refs per bom_plan (R41>R51, C38>C67, C40>C68, D9>D3, U2>U15, R45>R55, R42−44>R52−54, U18>U14).
Symbols from the OpenDrone catalogue with TI pin numbers: TLV7031 X2SON 1 OUT, 2 V+, 3 V− (centre pad), 4 IN−, 5 IN+; SN74LVC1G3157 DTB 1 B2, 2 GND, 3 B1, 4 A, 5 VCC, 6 S. The house lib symbols used the EasyEDA pad order (consistent with its own footprints), so the drawing is re−routed with the same nets: IN+ = VID_DC, IN− = GND, B1 = camera, B2 = OSD level. The house sheet's rotated ground is upright here.

Operation: OSD_EN low (RP2354A reset pull−down, OSD idle) −> B1, the camera passes through. OSD_EN high −> B2: OSD_W high = white, low = black. No OSD without camera sync (osd_pico.c overlays on detected sync). OSD_W/EN/SYNC = GPIO21/22/23, consecutive, PIO2 (PINMAP).

Sync threshold (P4−10): 680R/470p = 498 kHz: colour burst x0.11−0.14 (+−17−21 mV), 4.7 us sync kept (tau 0.32 us). C68 + D3 (cathode on VID_DC) clamp the sync tips to −VF; the leakage−limited clamp current (uA average, tens of uA at the tip) gives VF about 0.1−0.2 V over temperature (I: DS 0.37 V max at 10 mA / 25 C, 0.20 V at 10 mA / 125 C). IN− at GND puts the trip point VF above the tip, 33−67 % of the 0.3 V sync, >= 0.08 V from tip and blanking after VIO +−8 mV and VHYS 2−17 mV. OSD_SYNC is low during sync (osd_tx.pio waits for a low sync pin). IN+ at −VF stays inside abs max (VEE − 0.3 V), at most 0.1 V under the ROC input floor (VEE − 0.1 V), no phase reversal (DS 6.4.1). tpd 3 us on both edges (equal shift, as on the house board).

Ratings: +3.3V only (U14 1.65−5.5 V, U15 1.6−6.5 V). S: VIH 0.7 VCC = 2.31 V, edges <= 10 ns/V (fast slew pads). AC−coupled camera sync tips (−0.3 to −0.4 V) are below the switch ROC (0..VCC) but inside abs max (−0.5 V, 50 mA; 37.5R source), as on the house board; V9 checks. Supply: U15 0.9 uA, U14 <= 10 uA, divider 0.94−1.26 mA + OSD_W 2.74 mA. C68 X5R carries <= 0.6 V DC: no DC−bias loss (P4−13 hot−zone rule void, D69).

Placement (P5; P4−17 sheet note): bottom, region osd. BR−13: C70 is the only supply cap of U14 (VCC 5) and U15 (V+ 2): place it between the two parts (2.5 mm apart), GND via−in−pad; the P5 check lists its distance to both pins. U15 centre pad (3, V−) to GND through a via (TI 7.4). R51 and C66 at the video entry on the CAM pad J23 / plug J32 line; camera return into L2 at the CAM GND pad; video on L1/L6 over a plane with GND both sides, never L3 (spec 11). R55, C67, C68, D3 next to U15 IN+. Video parts away from the boost and its inductor (preference, D69; v2 floorplan 3.3−8.6 mm; V5 video test).
Netclass Analog: VIDEO_IN (flag here), VIDEO_OUT, VID_FILT, VID_DC, OSD_LVL, OSD_SYNC by the setup_board.py patterns; OSD_W, OSD_EN Default.
Sourcing: exact MPNs from bom_plan; LCSC empty where bom_plan has none (C66, R52, R54: open item A9−LCSC). PS−01 (R54 −> 270R RC0201FR−07270RL, shared with R59, −1 line): black 0.258 V, white 1.011 V, 187R, OSD_W 2.79 mA, also passes P4−10; proposed in contract−issues, drawn = bom_plan.

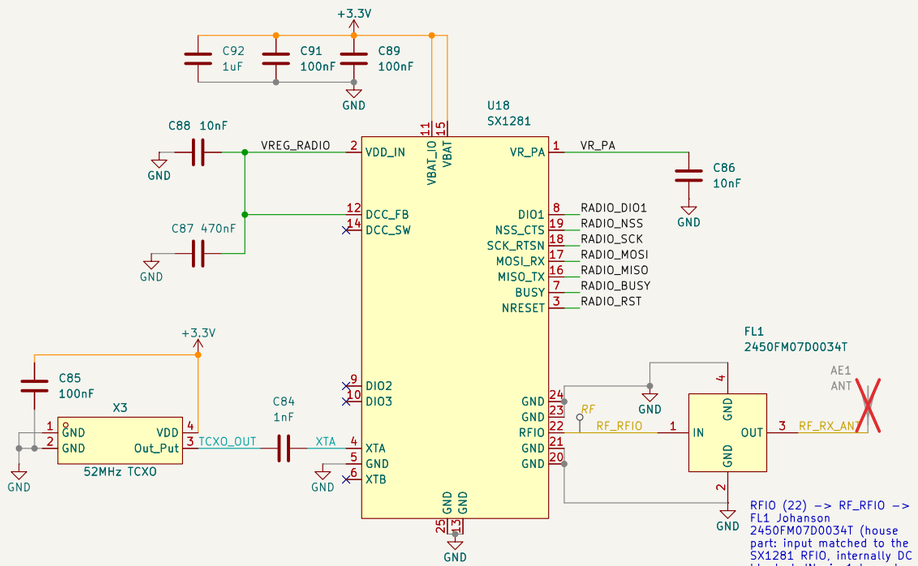
OpenDrone

Sheet: /OSD/
File: osd.kicad_sch

Title: PIO analog OSD front end

Size: A4	Date: 2026−10−09	**Rev: rev1**
KiCad E.D.A. 10.0.6		Id: 10/14

SX1281 RADIO + 2.4 GHz FILTER + WIRE ANTENNA



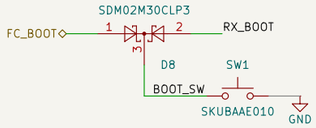
SX1281 (U18), copied from OpenRX-Lite-UFL; SX1281IMLTRT (D52 BR-25), SX1280IMLTRT drop-in alternate (ranging engine only).
LDO MODE (Semtech DS Rev 3.2 5.1, Fig. 15-4): DCC_SW (14) open, the sibling's 15 uH L4 deleted (the ELRS layout has no radio_dcdc); VDD_IN (2) tied to DCC_FB (12) = VREG_RADIO with C87 470 nF + C88 10 nF; VR_PA (1) C86 10 nF; VBAT_IO (11) C91 and VBAT (15) C89 100 nF (Table 15-1) + C92 1 uF radio local (P4-12 lever PS-06); the house sheet's extra 10 nF / 10 uF deleted (BR-06).
PIN 5 IS GND_DCDC (DS pin table) and is grounded here; the sibling sheets left it open (only XTB stays open with a TCXO, DS 15.2). DIO2 / DIO3 open.
52 MHz TCXO X3 (kept, D53): clipped sine AC-coupled by C84 1 nF into XTA (4), C85 100 nF at VCC, XTB (6) open.

RFIO (22) -> RF_RFIO -> FL1 Johanson 2450FM07D0034T (house part: input matched to the SX1281 RFIO, internally DC blocked, IN pin 1 toward the chip) -> RF_RX_ANT (50 Ohm) -> AE1: insulated lambda/4 wire (about 28-29 mm, trimmed in V7) in a plated DO.5 hole; AE1 DNP, out of BOM / CPL. RF class; board rule areas RF_RX_ANT / RF_RX_EXIT.

NOTES

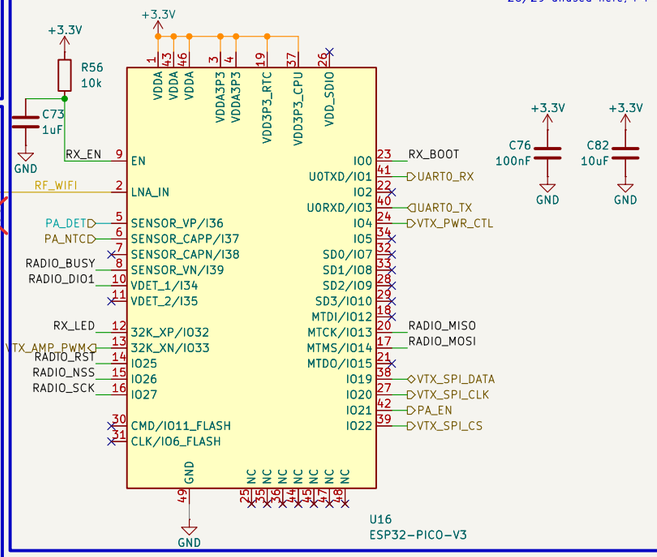
RX SHEET (P4 shard C). Sources: the SX1281 radio, 52 MHz TCXO, RGB LED and Wi-Fi pi match are COPIED from the house sheet OpenRX-Lite-UFL hardware/esp32c3_sx1281_lite.kicad_sch (same as OpenAIO rx_esp32c3_sx1281.kicad_sch): same symbol positions and wires, adapted to bom_plan refs/values/MPNs and to D52/D53/D58 (LDO mode, pin 5 to GND, U.FL replaced by the wire-antenna hole, 2450AT07A0100001T chip antenna, grounds turned down); the house Johanson 2450FM07D0034T front-end filter stays. The ESP32-PICO-V3 block is new, from the Espressif ESP32-PICO series datasheet v1.3 Fig. 11 (D52 BR-01, D55); the shared boot / bind button is new (D65, Matrix II style). Pin plan: research/PINMAP.md 6. Parts: hardware/bom_plan.json. Radio and LED +3.3V come from the POWER sheet (U4 LP5912-3.3).

BOOT / BIND BUTTON (D65)



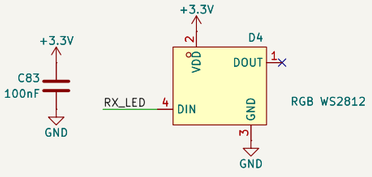
BOOT / BIND BUTTON (D65: new). SW1 Alps SKUBAAE010 (2.4 x 1.4 mm) from BOOT_SW to GND. D8 Diodes SDM02M30CLP3 dual Schottky, common cathode on BOOT_SW; A1 = FC_BOOT (RP2354A QSPI_SS through R44 1k, RP2350A sheet), A2 = RX_BOOT (ESP32 GPIO0, pin 23).
Released: each strap sees only diode leakage (IR <= 0.15 uA at 10 V, 25 C), so GPIO0 keeps its internal pull-up and QSPI_SS its own level.
Held while power is applied: RP2354A BOOTSEL and ESP32 ROM download mode (CSn <= 0.51 V vs VIL 0.8 V; GPIO0 <= 0.42 V vs VIL 0.825 V); each mode stays latched until that chip's next reset.
Production / ELRS recovery (D67, spec 12.1): (1) hold SW1 (or probe BOOT_SW) while powering; (2) write the Betaflight UF2 (or picotool reboot): the RP2354A restarts into Betaflight, the ESP32 stays in download mode; (3) esptool through Betaflight serial passthrough.
Pressed while running: ELRS button (bind 1.5 s, Wi-Fi 5 s, reboot 12 s); 2.9 mA through R44, CS stays >= 2.62 V, the FC keeps running.
Bench V-btn: if GPIO0 dips below 2.5 V while released (QSPI_SS edges under XIP load coupling through D8), add 100 pF 0201 on BOOT_SW.

ESP32-PICO-V3 MCU (ELRS)



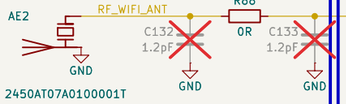
ESP32-PICO-V3 (U16), Espressif PICO datasheet v1.3 Fig. 11 / Table 4 (new block).
SUPPLY: VDDA (1, 43, 46), VDDA3P3 (3, 4), VDD3P3_RTC (19), VDD3P3_CPU (37) on +3.3V; C82 10 uF + C76 100 nF at VDDA3P3 pins 3/4 (Fig. 11 C1/C2). The 4 MB flash, 40 MHz crystal, CAP1/CAP2 network and the VDD_SDIO capacitor are inside the SiP (Fig. 8): VDD_SDIO (26) stays open.
EN: CHIP_PU (9) = RX_EN, R56 10k to +3.3V + C73 1 uF (tau 10 ms), never floating.
STRAPS at reset (P4-6, checked on the netlist):
GPIO0 (23) RX_BOOT: internal 45k pull-up, D8 blocks -> 1 = SPI boot (SW1 held -> download).
GPIO2 (22) and GPIO5 (34): no connect since the D111 map (internal pull-down / pull-up -> download-mode and SDIO timing straps at their defaults). PA_EN moved to GPIO21 (LP5907 EN 1 MOhm pull-down holds the PA off until ELRS drives it). VTX_SPI_CLK to GPIO20.
GPIO12 / MTDI (18): no connect, internal pull-down -> VDD_SDIO 3.3 V for the in-package flash (VDD_SDIO eFuse burn optional).
GPIO15 / MTDO (21): no connect, internal pull-up -> boot log on U0TXD (FC ignores it).
NOT CONNECTED: flash CMD/IO11 (30) and CLK/IO6 (31); NC pins 25, 35, 36, 44, 45, 47, 48; free IO7/8/9/10 (32, 33, 28, 29; PSRAM on the -02 alternate), IO2 (22), IO5 (34), I35 (11), I38 (7).
ELRS LAYOUT JSON (D53, D81: D111 map): serial_rx 3, serial_tx 1, radio_sck 27, radio_mosi 14, radio_miso 13, radio_nss 26, radio_rst 25, radio_busy 39, radio_dio1 34, led_rgb 32, button 0, vtx_nss 22, vtx_sck 20, vtx_mosi 19, vtx_miso -1, vtx_amp_pwm 33, vtx_amp_vpd 36, vtx_amp_vref 21. "vtx_miso": -1 is written explicitly: an overlay only replaces keys, so an omitted key keeps the stock 23 = the PICO flash DI, which SPIClass(HSPI).begin() turns into an input (flash cut, crash at VTX init); -1 makes the core attach the HSPI default GPIO12 (no connect here) instead.
patched ELRS: GPIO4 VTX_PWR_CTL (+3.3V_VTX LDO EN through R84 1k), GPIO37 PA_NTC (ADC1, read in the isTick window: erratum 3.11 pulls GPIO36/39 low for ~80 ns when SAR ADC1 powers up, and RADIO_BUSY is on GPIO39).
VTX LINES: series 1k R80/R81/R82 are drawn on the VTX sheet and placed at the ESP32 end (back-feed limit with +3.3V_VTX off, F21). GPIO33 VTX_AMP_PWM = ELRS vtx_amp_pwm (RTC6705 PAOUT1 feed PNP, not the SE5004L bias).
PA_DET is ADC1_CH0 on GPIO36 (readable with Wi-Fi on).
+3.3V LOAD: ELRS about 0.07 A in flight, Wi-Fi TX peaks 370 mA (DS Table 19, 802.11b 19.5 dBm) in ELRS Wi-Fi mode only (disarmed, bench); +3.3V peak about 0.48 A with the FC and LED, inside the LP5912 500 mA rating; LDO heat and the COUT sum belong to the POWER sheet (P4-12).
PLACEMENT: SiP site from floorplan v3 (D55, D61): no 85 C halo rule (D69). Alternate part ESP32-PICO-V3-02 (pins 28/29 unused here, P4-14).

ELRS RGB LED



D4 XINGLIGHT XL-1010RGBC-2812B (copied from OpenRX-Lite-UFL) on +3.3V, ELRS led_rgb GPIO32, one LED (overlay ledidx_rgb_vtx [0]).
P4-15 DECIDED: KEEP ON +3.3V. Datasheet: VDD 3.5-5.5 V, DIN VIH >= 2.8 V, 5 mA per colour with a 0.6 V sink knee. On +3.3V the supply is 0.2 V under the table minimum (blue / green may run below 5 mA: Vf + knee is about 3.4-3.6 V); DIN comes from a 3.3 V CMOS output on the same rail. On +5V the supply would be in range but the data guarantee is short (ESP32 VOH >= 0.8 VDD = 2.64 V at a high-impedance load, DS Table 18, vs VIH 2.8 V). Neither option sits fully inside the tables; the house RX circuit (OpenRX-Lite, -UFL, OpenAIO) runs this LED on 3.3 V, so it stays. Bench V-LED on the first boards: red, green, blue and white at 3.3 V, cold and hot. Fallback: VDD pin to +5V (one trace; C83 is 25 V).

Wi-Fi CHIP ANTENNA (D58)



Wi-Fi (D58): LNA_IN (2) -> RF_WIFI 50 Ohm line (<= 8 mm preferred, D69: FLOORPLAN open item 10) -> C133 shunt (DNP) / R88 0R series (fitted) / C132 shunt (DNP) -> AE2 Johanson 2450AT07A0100001T (FEED 1/4, GND 2/3). Same pi as Espressif Fig. 11 (C4/L1/C3) and OpenRX-Lite-UFL (C22/L2/C21). The PICO matches LNA_IN to 50 Ohm inside, so the network only tunes the antenna: values from V7 tuning (VNA at the feed or Johanson's service); R88 may become a Murata LQP03TN 0201 inductor. The AE2 footprint carries the 5.0 x 3.2 mm keep-out (DRU 'D21 wifi_antenna_edge_strip').

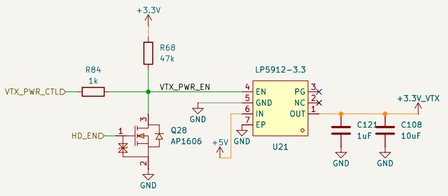
OpenDrone

Sheet: /RX/
File: rx_esp32_sx1281.kicad_sch

Title: ELRS 2.4 GHz RX: ESP32-PICO-V3 + SX1281

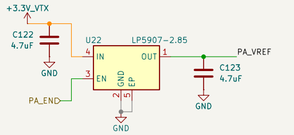
Size: A3	Date: 2026-10-09	Rev: rev1
KiCad E.D.A. 10.0.6		Id: 11/14

+3.3V_VTX LDO + HD GATE



U21 LP5912-3.3 on +5V: CIN = the shared 10 uF C13 on POWER (BR-14) only if P5 puts C13 within 1 cm of the U21 IN pin (LP5912 DS 9.2.2.2; tools/check_p5_conditions.py), else U21 gets its own 10 uF 0402 CIN (GRM155C80J106ME11D). EN = VTX_PWR_EN: R68 47k to +3.3V (stock ELRS runs the VTX): R84 1k from ESP32 GPIO21 (patched-ELRS power-down hook): GPIO4 low gives VEN 0.07 V < VEN(OFF) 0.3 V; Q28 AP1606 drain = HD gate (HD_EN high: <= 220 mOhm at VGS 2.5 V sinks 70 uA, 3.3 mA with GPIO4 high, VEN < 1 mV). Hot margin (P4 r4, R68 was 100k): EN stays >= VEN(ON) 1.3 V max up to 42 uA of Q28 leakage (100k: 20 uA): AP1606 IDSS <= 1 uA at 25 C, <= 100 uA at 125 C (VDS 20 V), about 16 uA at 85 C. HD_EN before Betaflight drives PINIO1: <= 0.22 V vs VGS(th) 0.4 V min at 25 C (250 uA), about 0.25 V at 85 C (~2.5 mV/K): a hot low-Vth part may keep the VTX off until PINIO1 is driven low (V9 hot-soak boot check). COUT on +3.3V_VTX: C108 10u + C121 1u + C122 4.7u + 3x 100n + 2x 100p = 16.0 uF nominal, about 6.8 uF effective at 3.3 V typical (5.4-8.2 uF at -/+20 %; POWER-sheet DC-bias figures), inside the LP5912 0.7-10 uF window. C108 sits at the RTC6705 VCO pins 39/40 (VCO bulk). Load 0.10 A typical, about 0.17 A at the top PA step (root power table).

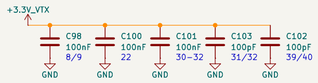
PA REFERENCE LP5907-2.85



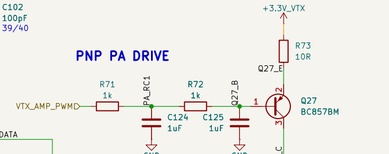
U22 LP5907-2.85 from +3.3V_VTX (TI SNVS798): HD mode removes PA_VREF with the rail. EN = PA_EN straight from ESP32 GPIO21 (D111 map: the LP5907 1 MOhm EN pull-down holds it off until ELRS drives it). C122 4.7 uF 0402 (> 0.7 uF effective), CIN = C122 4.7 uF 0402 X5R at U22 IN (about 2.6 uF at 3.3 V, >= 2 uF at -20 % and hot): LP5907 DS 5.6 / 7.2.2.4 needs > 0.7 uF effective over all conditions within 1 cm: the 1 uF G201 (about 0.45 uF) alone missed it (P4 critique r3).
DEVIATIONS, bench-gated: (a) VREF window: 2.85 V +-2 % = 2.79-2.91 V vs SE5004L VREF H 2.80-2.90 V, and the +-2 % is specified only for VIN >= VOUT + 1 V (VIN here 3.3 V): V5 measures PA_VREF on several units, cold and hot. (b) HD switch-on with ELRS holding PA_EN high: +3.3V_VTX collapses before C123 discharges, so VOUT exceeds VIN + 0.3 V (LP5907 abs max, DS 5.1) by up to one body-diode drop (<= 0.7 V, < 1 ms, <= 10 uJ from C123): V9 scopes PA_VREF and +3.3V_VTX (< 0.5 V). Fallbacks: C123 down to 1 uF, or the patched ELRS drops PA_EN on HD_EN.

RTC6705 VTX: 8 MHz CRYSTAL, SPI, VIDEO, PLL, PA DRIVE, PAOUT1 FEED

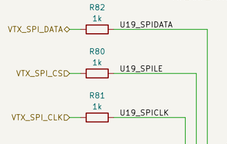
Decoupling at the RTC6705 supply pins (pin numbers next to each capacitor):



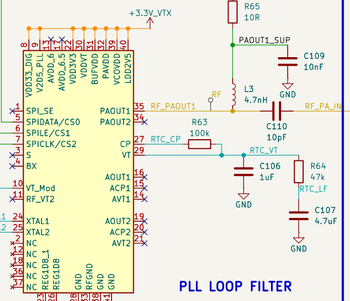
PNP PA DRIVE



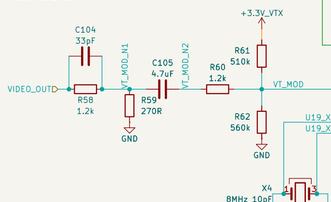
VTX SPI 1k SERIES



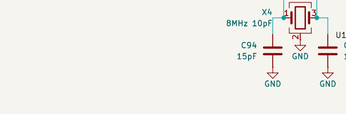
PAOUT1 FEED



VIDEO MODULATION NETWORK



PLL LOOP FILTER



RTC6705, checked pin by pin against OpenOSD-X v1.01: 23 REG1D8_1 -> C96 470n (= ref): 26 REG1D8 open (ref DNI); 17 AVDD_6.5 open (ref 3.3k + 1u, no 3.3 V: V5 checks for a 6.5 MHz spur): 31/32 BUFVDD/PAVDD -> +3.3V_VTX with C103 100p + C101 100n shared with 30: 40 LDO2V5 -> +3.3V_VTX direct with C102 100p (ref 51R + 1u filter; V5 video SNR): 39 VCOVDD with C102 + the 10 uF C108 placed there (ref 4.7u): 8/9 C98, 22 C100, Open: 1 SPI_SE (internal pull-up = SPI mode), 2-4, 11-16 (11 RF_VT2 audio input: V5 spur check), 18-21, 34, 36, 37, X4 exactly 8 MHz (ELRS R = 400), CL 10 pF = 15p + 15p in series + about 2.5 pF stray.

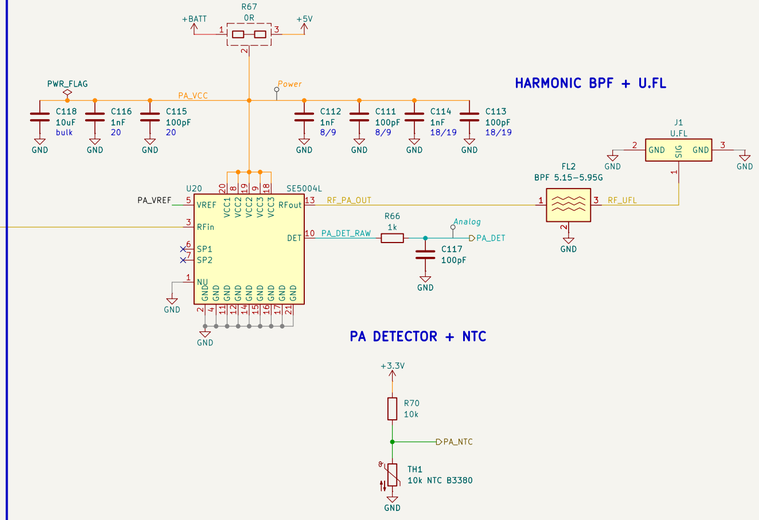
VTX SPI (PINMAP F21), R80 / R81 / R82 1k in series with CS / CLK / DATA, drawn here and PLACED AT THE ESP32 END of each line (bom_plan): they limit back-feed into the unpowered RTC6705 in HD mode (<= 2.7 mA per high line; CS idles high -> +3.3V_VTX about 0.27 V on the LP5912 100 Ohm active discharge, V9 limit 0.5 V; writes only at frequency changes). ELRS HSPI 1 MHz, mode 0, LSB first: tau = 1k x about 12 pF = 12 ns (26 ns 10-90 %) against >= 20 ns setup / hold in a 500 ns half period; the same RC on all three lines. SPIDATA is bidirectional (3-wire reads through R82, production test).

VIDEO + LOOP FILTER (OpenOSD-X v1.01). VIDEO_OUT (OSD switch: 75R camera termination on the OSD sheet) -> R58 1.2k || C104 33p -> R59 270R shunt -> C105 4.7u -> R60 1.2k -> VT_MOD, bias R61 510k / R62 560k from +3.3V_VTX (1.73 V) = OpenOSD-X R24 C17 R25 C21 R29 R30 R31. Load seen at VIDEO_OUT: 1.47 kOhm DC (R58 + R59), about 0.95 kOhm at 5 MHz and 0.60 kOhm at 10 MHz (R58 / C104 4.0 MHz, R59-loaded 21.9 MHz: 270R only above about 20 MHz, outside the video band): the OSD level note uses it. Loop filter = OpenOSD-X: CP -> R63 100k -> VT: VT -> C106 1u and R64 47k + C107 4.7u to GND. RTC_CP / RTC_VT / RTC_LF and VT_MOD* are high-impedance nodes (Analog net class): short copper, >= 5 mm from the boost and the ESC phases.

PNP PA DRIVE. ESP32 GPIO33 VTX_AMP_PWM (ELRS vtx_amp_pwm, 10 kHz) -> two RC poles (R71 / C124, R72 / C125, 1k / 1u each) -> BC857BM base; emitter R73 (Re) to +3.3V_VTX; collector Q27_C -> R65 10R -> PAOUT1_SUP -> L3 -> RTC6705 PAOUT1: it sets the RTC6705 output-stage current, not the SE5004L bias. Inverting, current mode. ngspice (Nexperia BC857BM model, PAOUT1 sink mode): Re 10R gives 32-59 mA at the top step over hFE 220-475 and -20..85 C, Q27 <= 114 mW (Ptot 130 mW at 85 C), base ripple 2.1 mV p-p, settles in 8 ms (ELRS 20 ms loop). GPIO33 after reset is an input (ESP32 DS IO_MUX: oe=0, ie=1, wpd: about 45 kOhm): ngspice case 4 gives 10-19 mA (Re 10R) from reset until ELRS Initialize() sets the pin count (stock vtxSPIPWM = MAX_PWM), and for as long as the ESP32 stays in ROM download mode (SW1 held, recovery): Q27 about 50 mW, the PA off (GPIO2 wpd). V5 measures that boot leakage at the U.FL. HD mode: both junctions reverse-biased 3.3 V (VEBO 5 V). Steps per 6 dB at Re 10R: 11-13 at 1 mA, 16-19 at 2 mA, 25-32 at 4 mA; V8 measures the PAOUT1 current at 25 mW and V5 sets Re (27R gives >= 20 steps from 1.5 mA). Start value R73 = 10R.
RATINGS AT THE TOP STEP (Re 10R, hFE 475, 85 C: 58.5 mA): R73 and R65 dissipate about 34 mW each, against the RC0201 derating (50 mW to 70 C, zero at 125 C: 36 mW at 85 C, 27 mW at 95 C) beside a 2 W PA: Q27 <= 114 mW against 130 mW at 85 C (12 % margin). V8 pass: top-step current and the board temperature at R73 / R65 / Q27 keep each part inside its derated rating; levers: cap the ELRS power-table top count so Ic <= 45 mA (P <= 20 mW), or Re 27R (26 mW in R73, 10 mW in R65) if V5 accepts the steps; 0402s only if V8 shows the top step is needed.

PAOUT1 FEED + DC BLOCK (OpenOSD-X L4 C40 C43 R34, fed by the PNP stage instead of its NPN follower). PAOUT1 (pin 35, +2 dBm) -> L3 4.7 nH choke to PAOUT1_SUP (C109 10 nF X7R) <- R65 10R <- Q27_C: C110 10 pF DC block into RF_PA_IN (SE5004L pin 3 'DC block required'). Only L3, C109 and C110 sit in the 5.8 GHz part-free band between pin 35 and the PA (50 Ohm CPWG < 5 mm, 45 degree). RF class on RF_PAOUT1 / RF_PA_IN / RF_PA_OUT / RF_UFL (pattern */RF_*).

SE5004L PA + SUPPLY, DETECTOR, BPF + U.FL



SE5004L PA (Skyworks 202393B + EK1). RFout has an internal DC shunt to GND and only RFin is marked 'DC block required': the BPF, U.FL and antenna carry no DC, so RFout -> FL2 directly (internal output match, no match parts on the line: P5 EM-checks the line). PA_VCC = R67 selector: one 0402 0R from COM to +BATT (default, pads 1-2) or +5V (pads 2-3, hand rework); the footprint anchor is the pad 1-2 midpoint, so the CPL centres the default part. VCC1/VCC2 <= VCC3 holds (one net). EK1 decoupling: 1n + 100p at 8/9, at 18/19 and at 20, on both package sides, + C118 10 uF shared, no ferrite. VREF from PA_VREF (abs max 3.6 V). SP1/SP2 (optional EVM caps) open: NU (open in the package) tied to GND: it lies in the DS ground metal. EP + pins 2, 4, 11, 12, 14-17 to GND with a via field.

PA DETECTOR + NTC (spec 4.8). DET (0.325 V no RF , 1.0 V at 27 dBm) -> PA_DET_RAW -> R66 1k / C117 100p -> PA_DET (ESP32 GPIO36, ADC1): the trace crosses sides, keep the RC at the ESP32 end of R66. TH1 NCP03XH103F05RL in the PA ring + R70 10k from +3.3V -> PA_NTC (GPIO37, ADC1, input only): 1.65 V at 25 C. BPF + U.FL: FL2 Walsin RFBPF1608060K9B0JC on the PA side away from the PA body (power handling at 0.71 W: open item O7): J1 U.FL with L2 cut out under the signal pad (RF_PAD_CUTOUT), >= 4 GND vias at the tabs.

SHEET NOTES

SHEET SOURCE. Generated by gen_vtx.py (hardware/tools/sch_build.py + sch_root.py). No OpenDrone sibling sheet or commons block exists for an RTC6705 VTX, so the circuit follows the OpenOSD-X v1.01 reference schematic (RTC6705 supply pins, video network, loop filter, PAOUT1 feed), the Skyworks SE5004L datasheet 202393B + EK1 and DESIGN-SPEC 4.8. Refs, values, MPNs and footprints: hardware/bom_plan.json. Signal flow: ESP32 controls (RX sheet) on the left, RTC6705 in the middle, PA, BPF and U.FL on the right.

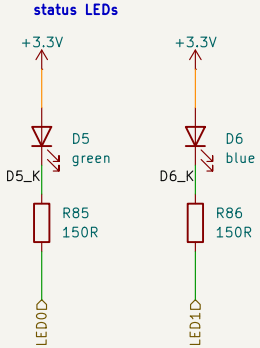
BENCH ITEMS on this sheet: RTC6705 pin 11 RF_VT2 open (V5 spur check): PA drive Re final value (V5 / V8, 27R fallback line): PA_VREF window and the HD switch-on excursion (V5 / V9, PA REFERENCE note): VIDEO_OUT OSD contrast and edges (V9, OSD sheet): X4 start-up margin (V5: series R 3-5x the 250R ESR, ~10 C cold start, slow supply ramp): Q27, R73, R65 temperature at the top step (V5 / V8, PNP PA DRIVE note).

OpenDrone

Sheet: /VTX/
File: vtx.kicad_sch

Title: 5.8 GHz VTX: RTC6705 + SE5004L PA

Size: A2	Date: 2026-10-09	Rev: rev1
KiCad E.D.A. 10.0.6		Id: 12/14



STATUS LEDS (D5 D6 R85 R86) source: OpenFC-Lite-Mini hardware/rp2350a.kicad_sch 'status LED' block (R52 + R53 2x 75R, D2 XL-1005UBC), copied and adapted: sink drive per hardware/sch_contract.json and research/PINMAP.md, 2x 75R -> one 150R 0201, second column for LED1.
Circuit: +3.3V -> LED -> 150R -> GPIO, the pin sinks (contract). Betaflight default polarity, low = on, no LEDx_INVERTED: LED0_PIN PA10 (GPIO10, green D5), LED1_PIN PA24 (GPIO24, blue D6).
Datasheet check (XINGLIGHT XL-1005UGC / UBC, 0402): VF 2.7-3.3 / 2.6-3.4 V at 20 mA, IF max 20 mA, PD 70 mW, VR 5 V. At a few mA the InGaN VF is about 0.2-0.3 V lower (inferred), so with VOL <= 0.5 V at 4 mA drive (RP2350 DS) the LED current is about 0.5-5 mA, typ. 2-3 mA, hard ceiling (3.3 - 2.4) / 150R = 6 mA: LED, 0201 resistor (<= 5.4 mW of 50 mW) and GPIO (4 mA setting, 100 mA IOVSS sum) all inside their ratings; brightness at 1-3 mA is about 17-135 mcd green and 4-36 mcd blue (Iv scaled from 350-900 / 80-240 mcd at 20 mA), enough for an indicator. Reverse bias is 0 V when the pin drives high (IOVDD = +3.3V).
Reset state: RP2350 pads come up with the pull-down on (PDE = 1, 36-113 kOhm), so both LEDs may glow faintly (about 10-30 uA) until Betaflight drives the pins and in BOOTSEL mode. Accepted, cosmetic.
Placement (P5): bottom side, fc_core region (bom_plan); the R may sit at either end of the LED, slow signal (D33). Pins LED0/LED1 = root nets /LED0, /LED1 (RP2350A).
P4 closed here: bom_plan R85/R86 'P4 checks brightness vs Vf' (above). P4-14 fields: RC0201FR-07150RL is a stocked Yageo RC_L part (Element14 4008725, RS 198-7568: 150R 1 % 50 mW 25 V); no LCSC number known, field left empty (contract open item A9-LCSC).

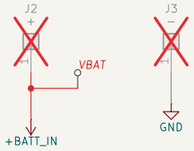
OpenDrone

Sheet: /LED/
File: led.kicad_sch

Title: Status LEDs

Size: A5	Date: 2026-10-09	**Rev: rev1**
KiCad E.D.A. 10.0.6		Id: 13/14

Battery pads (1S)



Source: OpenFC-Lite-Mini hardware/pads.kicad_sch (Conn_01x01 pad, pin down, rail symbol at the wire end).

J2 '+' = +BATT_IN, J3 '−' = GND: BT2.0 pigtail (22 AWG, user-fitted) through plated D1.1 holes in 3.0 x 2.0 mm pads (lib:BattPad_3.0x2.0mm_PTH1.1), >= 12 power vias each into the pours (spec 4.10, D72 power plan: +BATT_IN into L4 + L1/L6, GND into L2/L5). Silk '+' / '−' 2.0 mm and '1S' (spec 13).

1S LiPo/LiHV only, 3.0–4.35 V (D14). D1 SMF5.0A TVS, the pad bulk and the shunt R1 sit on POWER, which also holds the +BATT_IN and GND PWR_FLAGs. Netclass VBAT: directive here on +BATT_IN at J2.

Copper only: DNP, out of BOM and CPL (LINEUP A16).

Motor lands M1–M4: plug-ready, Molex PicoBlade header DNP



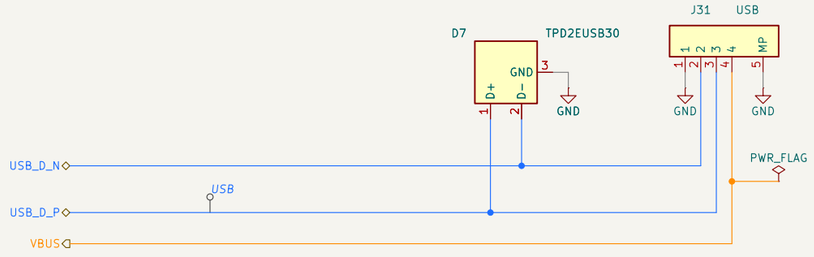
Source: new (D65: plug-ready lands as on the BetaFPV Matrix 1S 5IN1 II), drawn in the house pad style. One land per motor, lib:MotorPad_PicoBlade_1x03_P1.25mm_PTH0.5: pads 1 / 2 / 3 = Mn_C / Mn_B / Mn_A = /ESCn/PHASE_C / B / A

(symbol mirrored so A reads on top; phase order is free, Bluejay sets the direction). All four lands are placed on the top (F; bom_plan side = board), beside the corner ears (J4/J7 at one corner, J10/J13 at the opposite one), mouths facing out (D72).

Motor wires solder to the land (pads 1/3 1.65 x 1.50, pad 2 0.90 x 1.50) or through the 0.50 mm plated holes: the recommended default. The Molex 53047-0310 PicoBlade header (board side of the hobby 'JST1.25' motor plug, wire side 51021-0300) is a DNP BOM line, out of the CPL, hand-fitted from the far side with its pins trimmed on the land side. Molex rates 1.0 A per contact, −40..85 C, far below the motor current (4.3–5.2 A per channel, 12 A bursts): the published ESC ratings apply to soldered motor wires; the README pinout states the 1.0 A contact rating and that plugged motors run outside it at the user's risk (BetaFPV Matrix practice): V4 records the header temperature. Mating zone (D72): the plug faces outward over the ear neck, clear of parts.

Net names come from the ESC sheets (the root wires carry no label: /ESCn/PHASE_x). Netclass Phase: directive on M1_A here, the pattern /ESC?/PHASE_? assigns the rest. Phase copper runs from the AGM210MAP phase island straight into the land; solder areas 0.30 mm and phase copper 0.20 mm from the edge (SOLDER_PAD class). M1 rear right (ESC1), M2 front right (ESC2), M3 rear left (ESC3), M4 front left (ESC4): Betaflight quad-X.

USB: JST SH 1.0 mm 4-pin + ESD



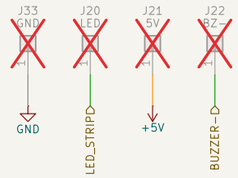
Source: OpenFC-Lite hardware/pads.kicad_sch (JST SH SM04B plug in a frame; there it carries a UART). The USB pinout, D7 and the VBUS flag are new (D57). Commons block USB_C_GCT4105_USBLC6 checked, not used (USB-C and a 6-pin ESD part).

J31 JST BM04B-SRSS-TB(LF)(SN) as in bom_plan (vertical, bottom): 1 GND, 2 D− (USB_D_N), 3 D+ (USB_D_P), 4 VBUS, tabs 5/6 GND = BetaFPV 'USB(GND DM DP 5V)' order (bom_plan, IO-SET row 6, LIBRARY D57). Pin 1 = GND is inferred: V-plan gate V0-USB before the P5 freeze (continuity on a BetaFPV SH1.0 USB adapter or a Matrix II close-up); a mirrored plug would put 5 V on board GND and −5 V on the SGM40661 IN (abs min −0.3 V). D76 (owner) keeps the vertical BM04B facing up next to a mounting hole (D72 side-entry SM04B withdrawn): its mating zone above the board stays clear of tall parts.

D7 TPD2EUSB30DRTR (TI SLVSAC2G): pin 1 D+, 2 D−, 3 GND (Table 5-1); VRWM 5.5 V, IO abs max 6 V, so a D+/D− short to VBUS survives; 8 V clamp at 1 A, 0.7 pF, IEC 61000-4-2 8 kV contact, −40..85 C. It sits at J31, ahead of the 27 R R42/R43 at the RP2354A (RP2350A sheet). Layout per DS 10.1: next to J31, protected traces straight with rounded corners, no unprotected trace beside the J31–D7 stretch, GND pin by via-in-pad into L2. USB pair 90 ohm, length-matched (D33); netclass USB from the directive on USB_D_P (_P/_N pair).

VBUS is a hierarchical net (LINEUP A11) to POWER: U24 SGM40661 OVP IN + C131 100 nF take the hot-plug and over-voltage duty (D57). PWR_FLAG here: J31 is the only source of VBUS. A cable with pins 1/4 swapped puts −5 V on VBUS (unprotected, as on the Matrix).

LED strip + buzzer row



Source: OpenFC-Lite-Mini hardware/pads.kicad_sch (LED-strip and buzzer pads).

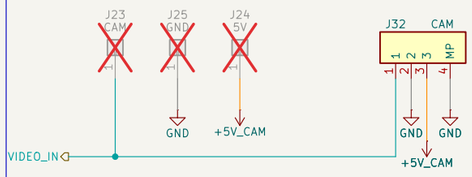
Row GND LED 5V BZ− (D63/D64): the strip gets its own GND pad J33 (supersedes D27).

LED = LED_STRIP driven straight from GPIO5 (PIO1 ws2812; D52 BR-04: no translator, no series resistor, Matrix II parity). A WS2812B on 5 V asks VIH 0.7 VDD = 3.5 V; the 3.3 V drive works on the common whoop strips, a level problem shows as a dead first LED (bench check, V9).

5V = +5V for the strip (0.20 A allowance, analog mode, spec 5.1) and the buzzer +. BZ− = BUZZER−: drain of the AP1606 low-side switch Q26, gate on GPIO25 (BEEPER_INVERTED, RP2350A sheet); the buzzer sits between 5V and BZ−.

Copper only (DNP, out of BOM and CPL, LINEUP A16).

Camera pads + JST SUR 0.8 mm plug



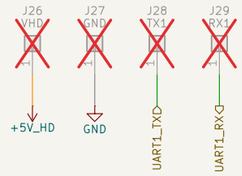
Source: OpenFC-Lite hardware/pads.kicad_sch (CAM frame: pads + SH1.0 SM03B plug), adapted to the JST SUR 0.8 mm SM03B-SURS-TF(LF)(SN) (D63/D64, IO-SET row 3: BetaFPV C03 and the Matrix-recommended cameras).

Plug and pads in parallel: J32 pin 1 VIDEO_IN, 2 GND, 3 +5V_CAM, tabs 4/5 GND; pads CAM / GND / 5V (J23 J25 J24). On the board the plug faces inward and the pads sit in front of its mouth: plug OR solder in one mating zone (D72).

VIDEO_IN -> OSD front end (TLV7031 sync + SN74LVC1G3157, OSD sheet), netclass Analog (OSD directive). +5V_CAM = +5V after FB1 120 R + C15 10 uF (POWER), 0.12 A. Video return: CAM GND pad and plug pin 2 enter L2 at the CAM pad (spec 11).

Gate G1 still open: pin order provisional until calipers + photo on a C03 lead and the MX2 CAM IN. With GND in the middle a mirrored footprint puts 5 V on the camera's video output. G1 = 1.0 mm pitch -> SM03B-SRSS-TB(LF)(SN), same pins.

HD pads (DJI O4 Lite)



Source: OpenFC-Lite-Mini hardware/pads.kicad_sch (digital video pads), new pad set (D13/D16).

VHD = +5V_HD from U5 TPS22810 (on with HD_EN and cell >= 3.23–3.82 V, held down to 2.74–3.14 V; POWER P4-1, 1 % resistors), 1.20 A budget; O4 Lite input 3.7–13.2 V. TX1 = UART1_TX (GPIO16, hardware UART0) -> O4 RX, RX1 = UART1_RX (GPIO17) <- O4 TX: MSP DisplayPort; no SBUS pad (D16). Cut DJI cable: red -> VHD, black + brown -> GND, white -> TX1, grey -> RX1, yellow open (no SBUS input since D89).

Analog builds: TX1/RX1 are the free hardware UART (the only one since D89, no user UART pads), or I2C0 via the I2C-HD preset (IO-SET row 8).

P4-11: the O4 stays unpowered until HD_EN, so its TX cannot drive RX1 before IOVDD (FT pads limited to 3.63 V unpowered, PINMAP F16). A device on TX1/RX1 in an analog build runs from a 5V pad (LED row), which comes up before +3.3V: 3.3 V logic only (a 5 V TX on RX1 can exceed 3.63 V during the ramp).

Copper only (DNP, out of BOM and CPL, LINEUP A16).

Mounting holes



H1–H3: the 25.5 mm M1.4 pattern (D73) is cut in Edge.Cuts (ear holes with flange keep-outs, spec 7), so the symbols are excluded from the board (FLOORPLAN 'For P4'), BOM and CPL; footprint kept as reference. No net (non-plated; GND annulus = 016). LOGO1 (incutec, B.Silkscreen) is a board-only footprint, no symbol.

PADS: every off-board connection. Style from the OpenFC-Lite-Mini and OpenFC-Lite pads sheets (pads pin down, a rail symbol or a hierarchical label at the wire end, plug blocks in frames); the source of each block is in its note. References, values (pad codes = silk codes), footprints, MPN / Manufacturer / LCSC and the DNP / BOM / CPL attributes are hardware/bom_plan.json; solder pads are DNP and out of BOM and CPL (LINEUP A16), the motor header is a DNP BOM line.
20 hierarchical labels = the PADS sheet pins (sch_contract.json): VIDEO_IN, M1_A..M4_C, UART1_TX/RX, LED_STRIP, BUZZER−, USB_D_P/N, VBUS. Rails by power symbol (+BATT_IN, +5V, +5V_CAM, +5V_HD, GND; sources on POWER); VBUS carries its PWR_FLAG here. Netclass directives: VBAT (+BATT_IN at J2), Phase (M1_A), USB (USB_D_P). No user UART group (D89): the free UART is the HD group TX1/RX1 (UART1).
Owed: J31 pin 1 = GND (gate V0-USB before the P5 freeze), J32 pin order (gate G1), motor-header temperature (V4).

OpenDrone

Sheet: /PADS/
File: pads.kicad_sch

Title: Pads, motor lands, camera, HD, USB

Size: A3	Date: 2026-10-09	**Rev: rev1**
KiCad E.D.A. 10.0.6		Id: 14/14